



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computer Science
and Information
Systems

UNIVERSITI TEKNOLOGI MALAYSIA, JOHOR BAHRU

FACULTY OF COMPUTING

SEMESTER II - 2014/2015

SCSR1213 – NETWORK COMMUNICATIONS

GROUP PROJECT: CASE STUDY

GROUP NAME: *Fantastic 4*

GROUP MEMBER:

- | | |
|---|-------------|
| 1) MOHAMAD FAKHRULRADHI BIN MOHD SHA'RI | (A14CS0167) |
| 2) TIFFANNY ANAK GARAN | (A14CS0182) |
| 3) SYAKIR ZUFAYRI BIN ROSLAN | (A14CS0139) |
| 4) NATALIE NUAI ANAK GERALD | (A14CS0085) |

LECTURER: EN. MUHALIM BIN MOHAMED AMIN

DEDICATION AND ACKNOWLEDGEMENT

This project is dedicated to our Network Communications lecturer, En Muhalim Bin Mohamed Amin, who taught us a lot of things and tips for this subject and also in real life. Our beloved lecturer has shared a lot of things and also opinions with us to make sure our group project become a success. We have learn a lot of things when doing this group project that is given by our lecturer such as making a floorplan, planning budget properly in a project, making a topology, think logically and many more.

We want to say thank you so much to our lecturer. We have enjoyed the opportunity to watch and learn a lot from his knowledge and experience. We hope that we can pass our final exam and get A+ with flying colours.

We also want to say thank you to our family and friends who kept supporting us and shared with us a lot of ideas for our group project. They helped us a lot in sharing theirs ideas and also doing group project together at the faculty. We work hard together to make sure that we can done the group project perfectly before the dateline.

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1.0 GROUP INFO

Section: 08

Group Name: *Fantastic 4*

Group Number: 05

Group Member:

1) MOHAMAD FAKHRULRADHI BIN MOHD SHA'RI

Matric No.: A14CS0167

Program: 1SCSR

2) TIFFANNY ANAK GARAN

Matric No.: A14CS0182

Program: 1SCSV

3) SYAKIR ZUFAYRI BIN ROSLAN

Matric No.: A14CS0139

Program: 1SCSV

4) NATALIE NUI ANAK GERALD

Matric No.: A14CS0085

Program: 1SCSV

Budget Given: RM500,000.00

Meeting Info:

Meeting #1

Date: 30/03/2015

Time: 9:15PM

Place: In front of Dewan Serbaguna (DSG), Kolej Tun Dr. Ismail (KTDI).

Duration: 3 hours

Meeting #2

Date: 17/4/2015

Time: 3:00PM

Place: In front of Dewan Serbaguna (DSG), Kolej Tun Dr. Ismail (KTDI).

Duration: 3 hours

Meeting #3

Date: 03/05/2015, 07/05/2015

Time: 9:15PM

Place 1: In front of Dewan Serbaguna (DSG), Kolej Tun Dr. Ismail (KTDI).

Place 2: Foyer, N28a

Duration: 3 hours, 7 hours

Meeting #4

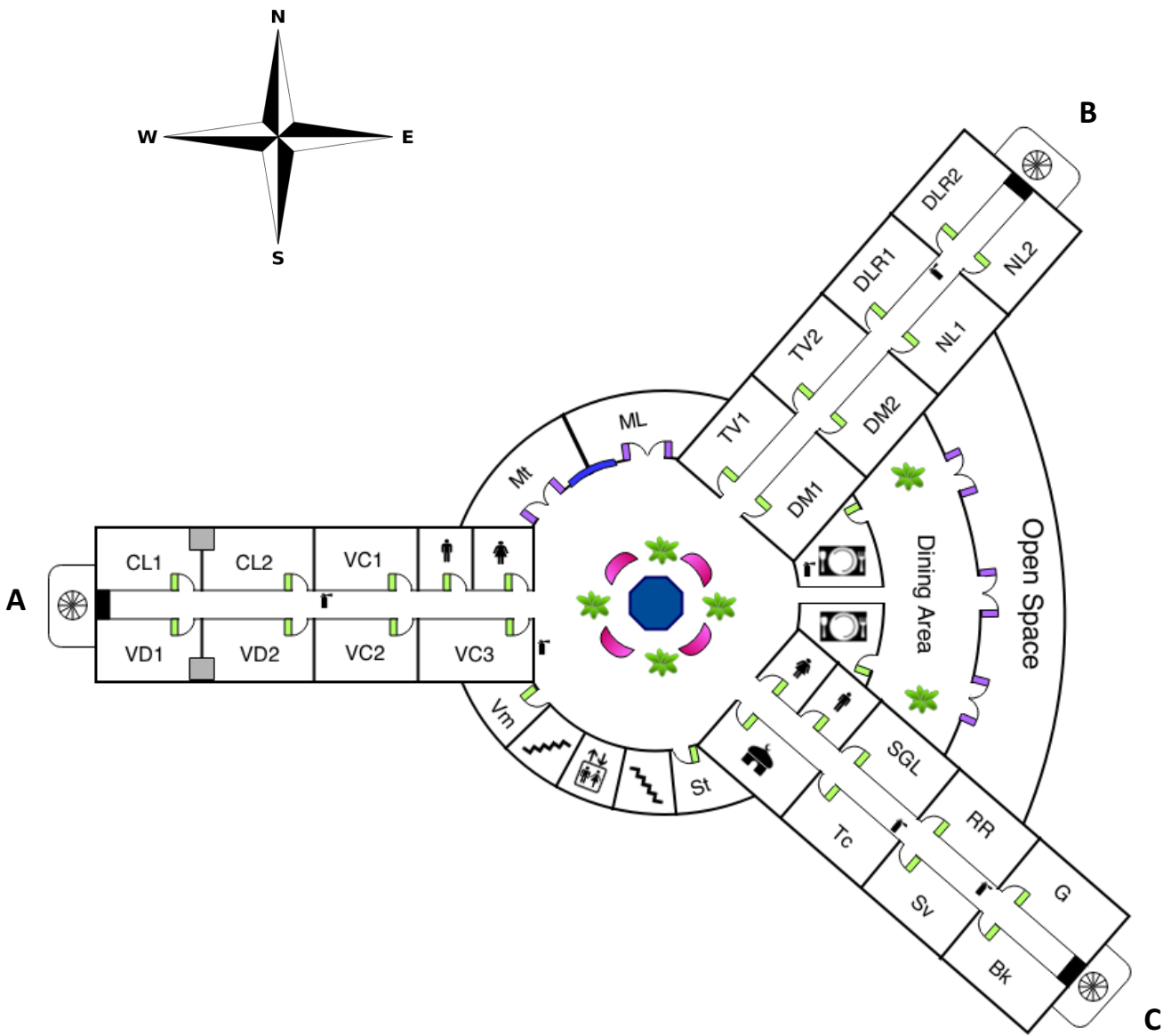
Date: 19/05/2015, 26/05/2015

Time: 9:30PM

Place: Foyer N28

Total Duration: 6 hours

2.0 FLOOR PLAN



Symbol List:

Sofa

CL1 : Computer Lab 1



Aquarium

CL2 : Computer Lab 2



Plant

VD1 : Video Editing 1



Back Door

VD2 : Video Editing 2



Notice Board

TV1 : TV Production Studio 1



Door

TV2 : TV Production Studio 2



Door

Mt : Meeting Room



Double Door

ML : Mini Library



Cafeteria

VC1 : Video Conferencing 1



Elevator

VC2 : Video Conferencing 2



Toilet (Women)

VC3 : Video Conferencing 3



Toilet (Men)

DM1 : Digital Media Lab 1



Surau

DM2 : Digital Media Lab 2



Stairs

NL1 : Networking Lab 1



Spiral Stairs

NL2 : Networking Lab 2



Fire Extinguisher

DLR1 : Distance Learning Room 1

Vm

Vending Machine

DLR2 : Distance Learning Room 2

St : Storage Room

SGL : Student General Lab

RR : Reserve Room

G : Generator Room

Tc : Technician Room

Sv : Server Room

Bk : Backup Room

Floor Plan Explanation:

We were asked to design our own floor plan. So, we have decided to list down the rooms which meet the requirement and also the room that we think it's necessary. Firstly, our building shape is a torchlight style shape.

This floor can hold about more than 400 people, including the visitors as well as the staff and administrator, and is provided with Wi-Fi hotspot in almost every area in this floor.

So here, in our floor design, we divided it into three wings that surrounded by a variety of rooms. At the west side, we have:

1. Two Computer Labs
2. Two Video Editing Labs
3. Three Video Conferencing Rooms
4. Toilet

At the east side, we have:

1. Two TV Production Studios
2. Two Distance Learning Rooms
3. Two Digital Media Labs
4. Two Networking Labs
5. Student General Lab
6. Technician Room
7. Reserved Room
8. Server Room
9. Generator Room
10. Backup Room

As mentioned earlier in the requirement, we need to use fiber optic cable for Digital Media Labs, Distance Learning Rooms, TV Production Studios and Networking Labs which are from the east side, according to the case study. We know that the price of fiber optic cable is a bit expensive. To reduce our cost, we put all the rooms in the same wing. At the same time, the speed and the quality remain the best.

A Server Room is located near to the east side because that side will be using most of the cables. As the fiber optic cable is more expensive than other cable, we decided to put it nearer to that side. In case there is troubleshoot or blackout in this floor, we also provide a Backup Room. The Generator Room is use for standby power supply. Besides that, a reserved room is use for the students or staffs for any program related to faculty or as a new room in the future.

Besides that, we also have an elevator between stairs. We can use stairs anytime that we want, but some prefer use elevator more. But anyhow, stairs are really important and are located on both sides of the elevator. Other than that, every wing has a spiral stairs at the end of it for emergencies purpose such as if building caught in a fire.

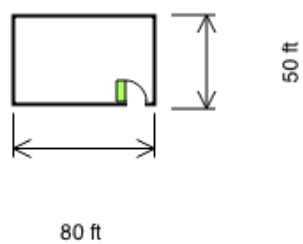
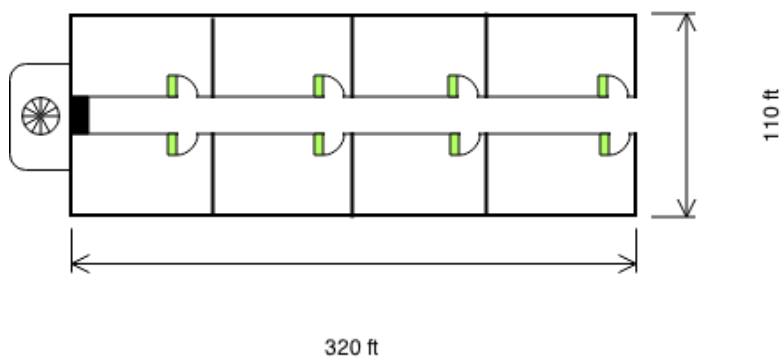
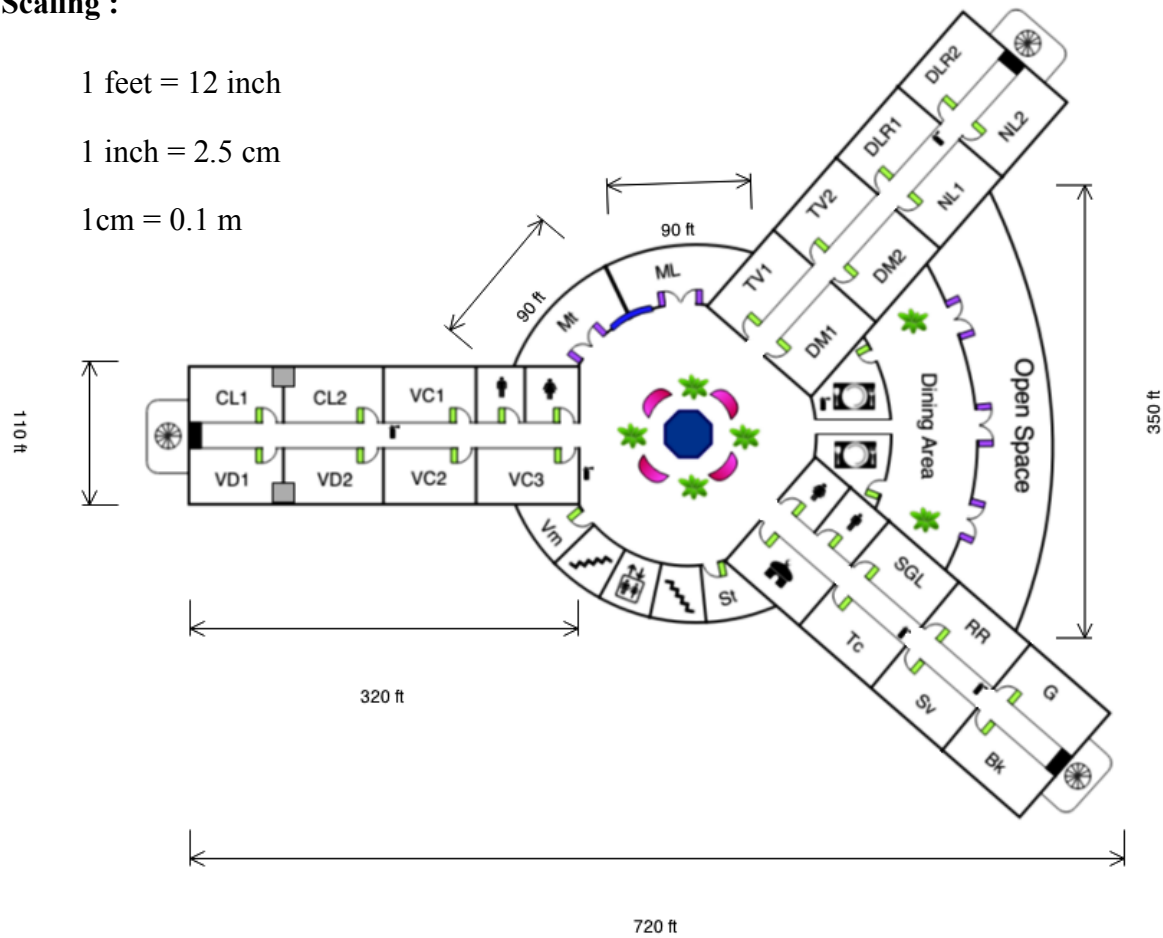
We also have a mini library and a meeting room that is located on the north. The vending machine and the storage room is located on the south. The storage room is a dedicated room for storing cleaning equipment for the cleaner. Besides that, there are also cafés for students and staffs to have their breakfast and lunch. At the cafés, we also have an open space, which students and staffs can have their breakfast and lunch there or just need to relax. At the center of the floor, we also have an aquarium and many sofas around it for students to relax, meet their friends, study or do their discussion.

Scaling :

1 feet = 12 inch

1 inch = 2.5 cm

1cm = 0.1 m



3.0 PRELIMINARY ANALYSIS

After discussed with each other, we had identified the questions that we need to ask in order to get the information necessary to develop a network plan for the case study.

Question #1: What is a router, modem, access point, and switch?

Answer:

- 1) A *router* is a device that moves data packets along networks. Routers are **located at gateways**, the places where two or more networks connect. Routers are typically used in **home networks** to share a single internet connection with multiple computers. A router is connected to at least two networks, commonly two LANs or WANs or a LAN and its ISP's network.

Initial routers provide **a few Ethernet ports** for wired connections. Nowadays, modern routers also provide wireless connectivity too. A router when connected, can provide access to a local network (LAN), but it does not certainly provide access to the internet. **In order for devices on the network to connect to the internet, the router must be connected to a modem.** Hence, most routers have a specific Ethernet port that is designed to connect to the DSL modem or Ethernet port of a cable.

- 2) A *switch* serves as a controller, allowing networked devices to talk to each other efficiently. **Switches create a network.** Routers connect networks. A router links computers to the Internet, so users can share the connection. A router acts as a dispatcher, choosing the best path for information to travel so it is received quickly. Through information sharing and resource allocation, switches save businesses money and increase employee productivity.
- 3) A *modem* is a connection device that **provides access to the Internet.** The modem connects to your ISP, which commonly provides either DSL Internet service or cable. Cable modems have a coaxial (or "coax") connection, which is the same type of connector found on a TV or cable box. This connects to a cable port on the wall. DSL modems have

a telephone connector (RJ-11 jack), which can be used to connect to a telephone socket on the wall.

By connecting your modem to your router, **all devices connected to the router can access the modem, and, therefore, the Internet.** The **router provides a local IP address to each connected device**, but they will all have the same external IP address, which is assigned by your ISP.

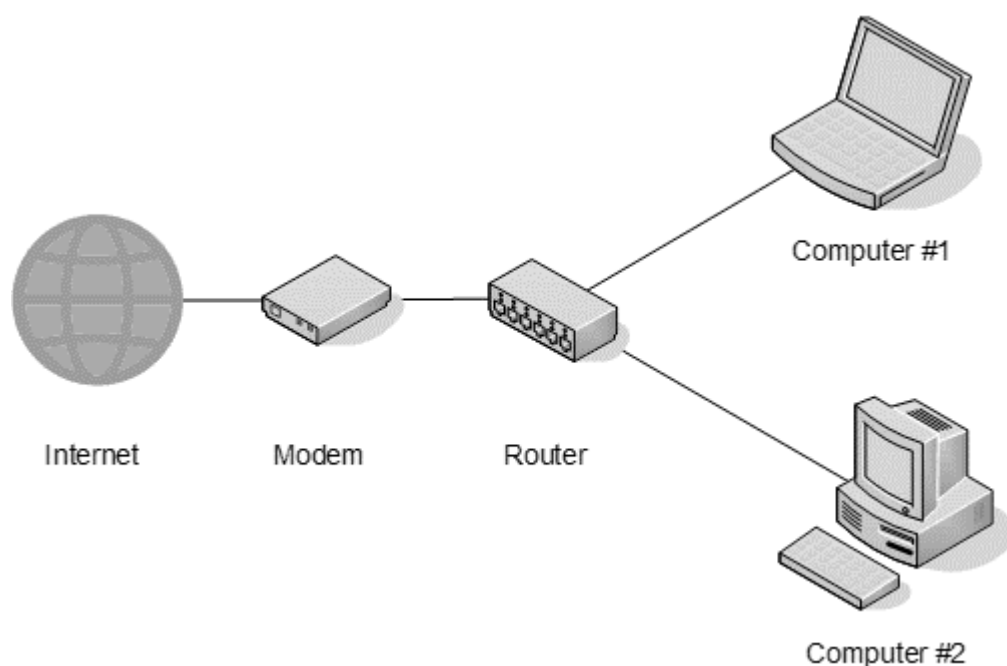


Figure 1: Device Connection (Source:

<http://www.microsoft.com/library/media/1033/windowsxp/images/using/networking/setup/68573-router-diagram.gif>)

In some cases, while the router and modem are usually separate devices, the modem and router may be combined into a single device. This type of hybrid device is sometimes provided by ISPs to ease the setup process.

- 4) An *Access point*, which is also known as a **base station or wireless router**. It is the location of a wireless receiver that enables a user with wireless access to connect to a network or the Internet. This term can refer to both Wi-Fi and Bluetooth devices.

Question #2: How to create a Local Area Network (LAN) that can connect to the internet?

Answer:

The devices connection order can be summaries as below:

1. PC or any wireless devices
2. Router
3. Modem
4. ISP (Internet Service Provider)
5. Internet

Some example of simple diagrams (Both have access to the internet):

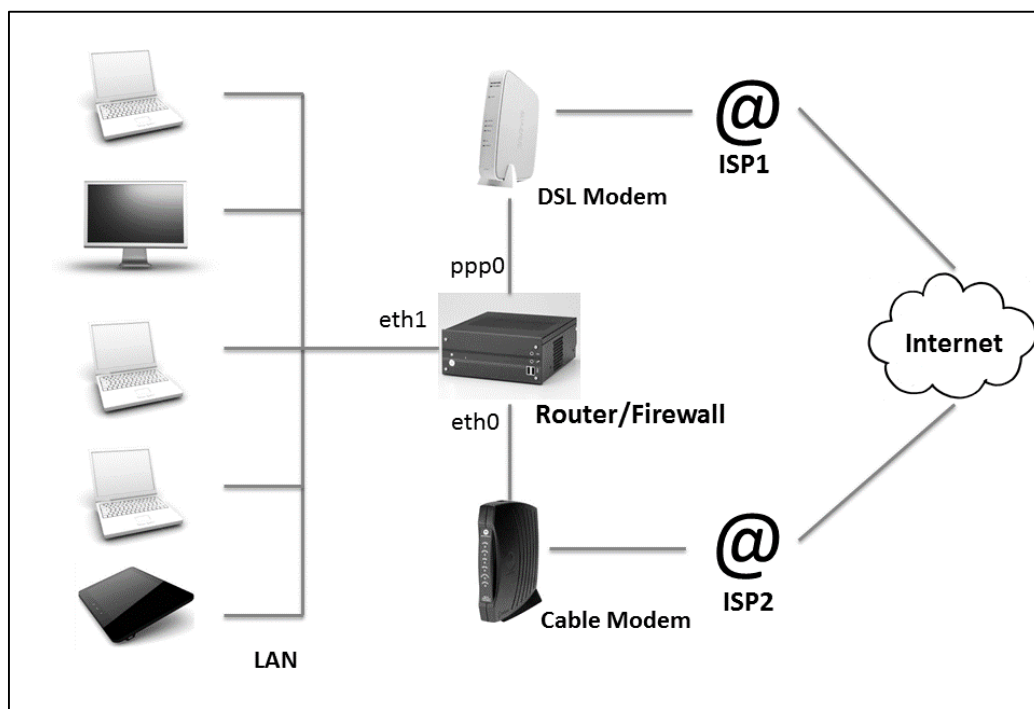


Figure 2: LAN Setup (Source: <http://lstein.github.io/Net-ISP-Balance/images/network-topology.png>)

Figure 3: LAN Setup (Source: <http://www.computer-networking-success.com/images/home-wireless.jpg>)



Firstly, we need to consider which type of LAN topology that we want to implement in our network. It is important to consider what will happen if one of the cables fails. In ring, line and bus topologies, a failure in a cable can affect many dependent computers to lose a connection. Hence, star topology is much more common and reliable, but it requires more cable. The dot at the center of the star is either a server or a sharing information between the other computers shown. Tree and mesh topology are choices that can be considered as well.

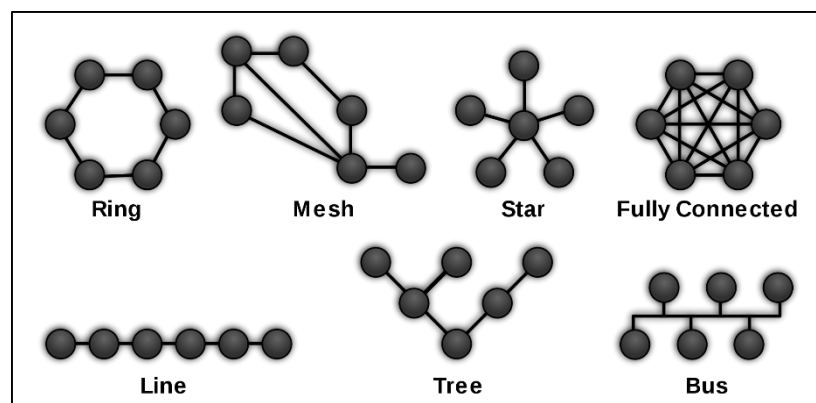


Figure 4: Network Topology (Source: <http://upload.wikimedia.org/wikipedia/commons/thumb/9/97/NetworkTopologies.svg/2000px-NetworkTopologies.svg.png>)

Next, we will need to prepare networking equipment that is required to set up a computer network such as:

- a. **Router / Wireless Router** (Access Point)
- b. **Cable** (If not wireless)
- c. **Switch** (Useful for star, tree and mesh network topologies)
- d. **Cards and adapters** (Each computer/devices need to have this hardware to connect to the LAN)
- e. **Server** (Most small networks do not need one)
- f. **Shared Hardware** (Example: Printer, scanner)

Here are the **steps** to set up a LAN that have access to the internet.

1. Install a router.
 - a. Using a cable (such as CAT-5) to connect to the internet source (ISP).
2. Connect wired computers to the router.
 - a. If the router has enough ports for all of the computers on the network, we can attach them directly using cable.
 - b. If the router does not have enough ports, move on to step 3.
 - c. Can act as a center for a star topology.
3. Connect a switch to the router.
 - a. Use a switch if the router does not have enough ports for all the computers on the network, or if a group of computers is in another room.
 - b. The switch will be connected to one of the ports on the router via a cable.
 - c. A star network will be formed when the computers connected to the switch.
4. Connect wired computer to the router.
5. Connect a wireless access point to the router.
 - a. If a more additional connection is required, provide a wireless access point in the network.

Question #3: What are the function of a Backup Room and Generator Room? Do we need to consider these rooms as important too?

Answer:

ICT rooms need to be ideally located in the building. Besides the size of the rooms, we need to consider aspects related to **cable security, fire resistance, noise, heating, electrical fields, conduit paths, equipment transport, floor loads** and any **general structure** of the building that likely can affects those rooms.

1) Backup Room

The Backup Room can provide a recovery point if suddenly primary ICT systems fail (such as virus attack). Any backup systems should be, as far as possible, fully independent of the supporting infrastructure used for the primary system so that in case of a failure of the primary ICT system, the secondary ICT system does not also fail. Backup ICT room should be frequently checked to ensure their continued operation. One of a biggest threat such as hacker or virus attack can be minimised by having a backup room in the institution.

Backup room is important because if anything happens to the server, which resulting many data become corrupted because of threats (either outside or inside), we still can cover the damage by using the data that have been 'backup'ed earlier. Usually, we will need a backup room for each server room.

2) Generator Room

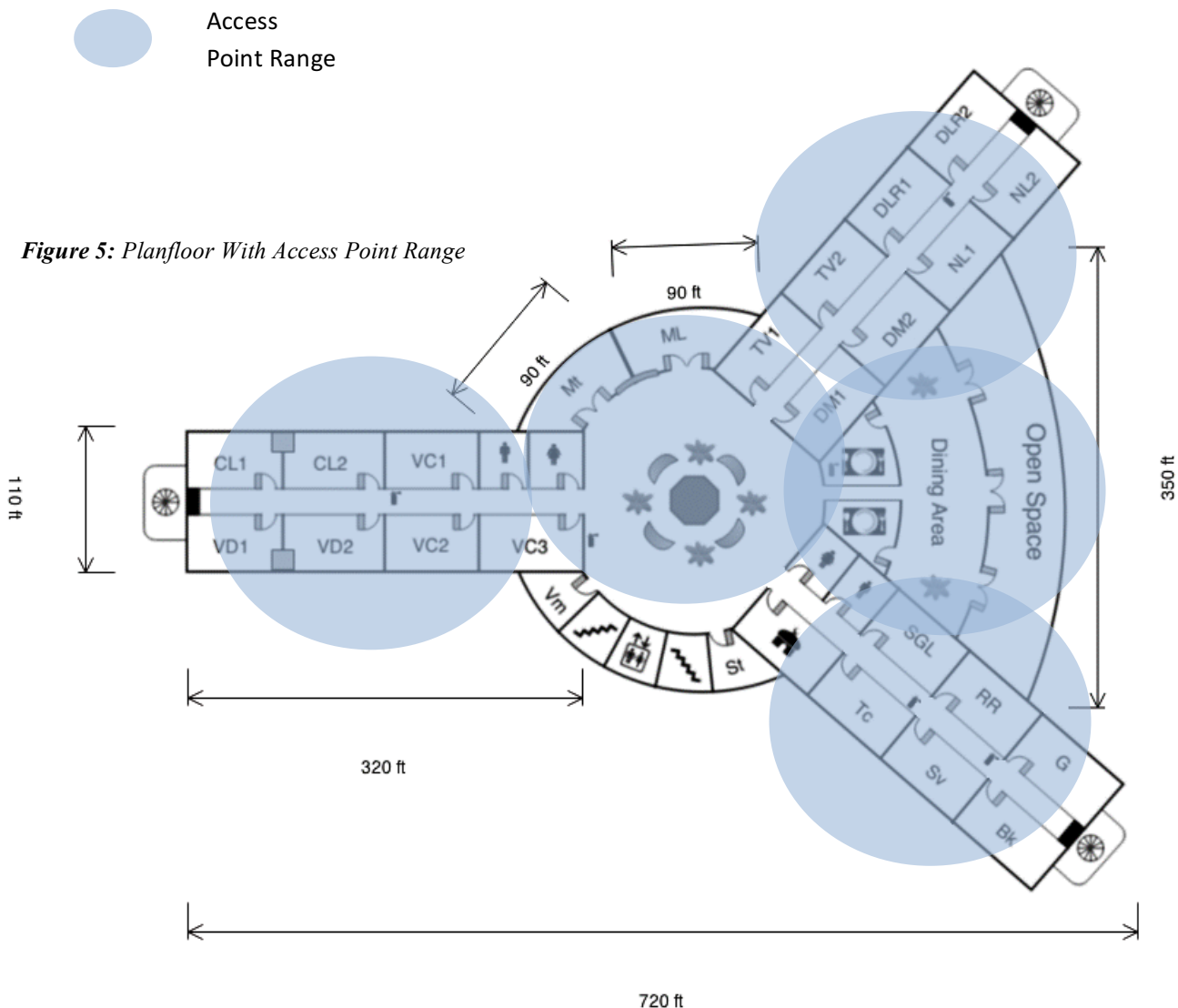
This room shall house a generator for the production of standby power. In this room, the generators sit very still and quiet, waiting for their moment to come to life and provide the emergency power they were designed for. The generators have to be in a separate room dedicated for the purpose of the room.

One of the important room too because if blackout happens, we still can provide emergency electric supply to keep the server in the Server Room ongoing for the user to access the server as frequently as possible.

Question #4: How to ensure that every user can access the wireless internet connection?

Answer:

Wired and wireless are both important to ensure that the internet connection can be access by all user. Assume that, we can provide about five (5) access point (wireless router) to make sure that the students and also the staffs can have access to the internet almost anywhere. For now, below is the location of the access point in our floor plan. Cable will be used in each important room that need higher bandwidth and more reliable connection. *(Assume that if we use an access point that support a range of up to 150 feet (46 m) to 300 feet (92 m).)*



Question #5: Which equipment that requires protection and what kind of security measure that we need to implement in a building?

Answer:

ICT equipment that **requires protection** includes any device which can store information electronically, such as:

- a. Computers (desktop, laptop or tablet)
- b. Photocopiers and printers
- c. Fax machines
- d. Mobile telephones
- e. Digital cameras
- f. Personal electronic devices

The **level of protection** that should be given to equipment is based on the higher of business impact level that would result from the compromise, loss of integrity or unavailability of the combined electronic information held on the equipment, or the loss/unavailability of the ICT equipment itself. ICT system equipment that needs physical security includes:

- a. Servers, including dedicated devices and laptops used as servers.
- b. The supporting network infrastructure (Example: cabling, patch panels)
- c. Gateway devices, for example, routers and network access devices.

ICT equipment **may require a controlled atmosphere** to ensure the safety of the information held on the equipment. ICT equipment holding information may also need a controlled environment to prevent failure of the equipment and potential loss of information. This may include, but not limited to, controlling:

- a. Temperature
- b. Humidity
- c. Air quality (smoke and dust)
- d. Water
- e. Light.

After discussing with each other about how to improve the security of the building and ‘ready for anything’ situation, we have decided that **fire extinguisher, CCTV, and periodic checks** are very important in this case study project.

1. Fire Extinguishers

- a. Have been included in the floor plan
- b. More fire extinguishers are placed near the Server Room if suddenly fire occur near the room.

2. CCTV (Closed-circuit television)

- a. Most important place that need a CCTV is near the Server Room and the lift.
- b. Important room such as Server Room, Backup Room and Technician Room will be put with CCTV in it too.



Figure 6: CCTV
(Source: <http://www.mantrat ec.com/Dome-cameras/dome-camera-1.jpg>)

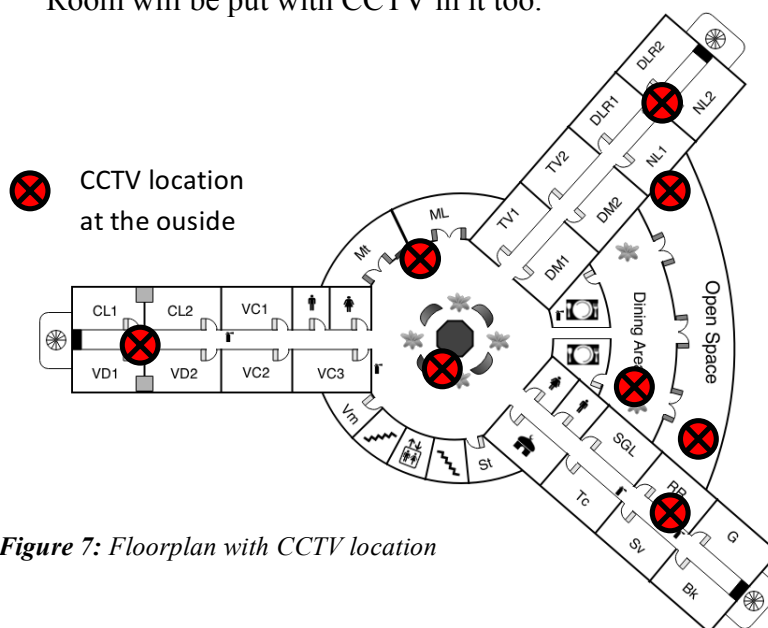


Figure 7: Floorplan with CCTV location

3. Periodic Checks and Speaker

- a. Done by the technicians.
- b. Will alert everyone if anything happens such as intruder bypass the floor security or fire occur by using the speaker.
- c. Ensure that Generator Room, Backup Room, and Server Room are always working and can be used without any problem.

4.0 DETERMINE FEASIBILITY

After our group has come to some conclusions about the requirements, then the feasibility of the project has been determined. So, our group has enough information or ideas to move forward with our own design of the floor plan.

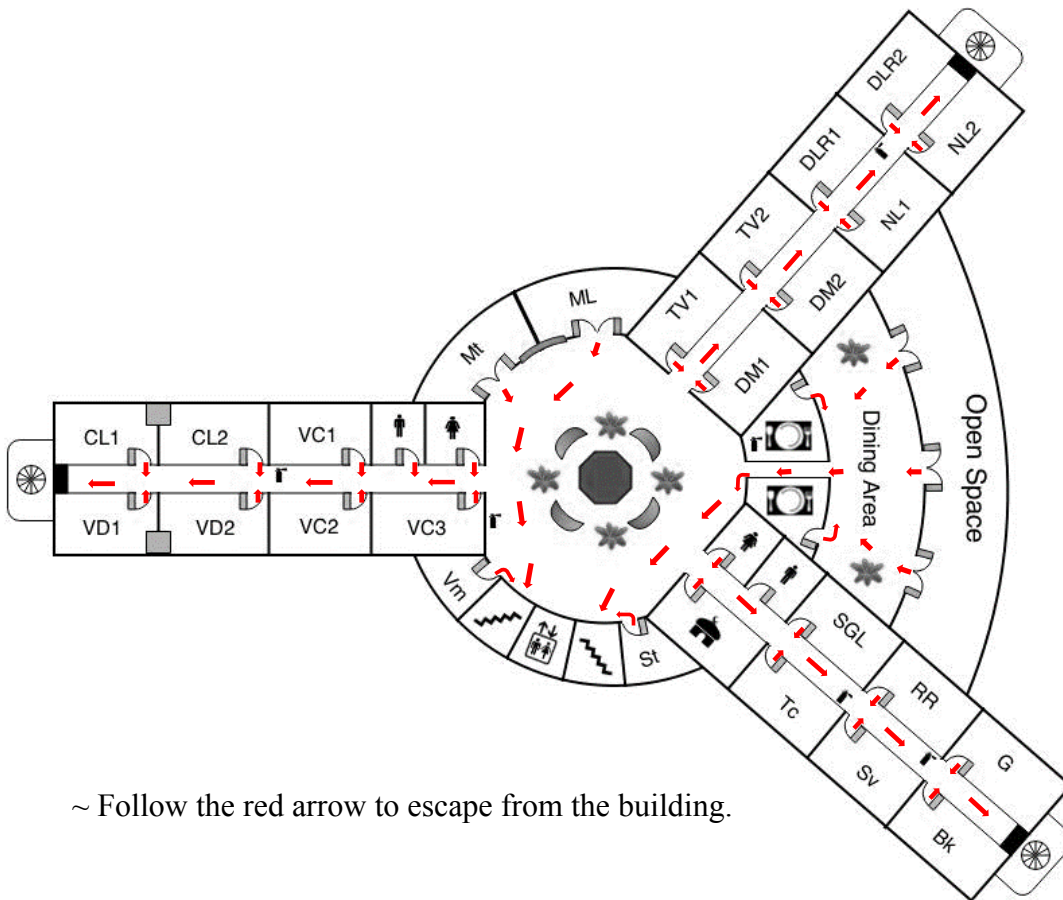
Based on the floor plan that we had done, the budget given which is RM500,000, it is enough and possible to move on with our project. The reason is because we expect that with the budget given, we try to reduce the cost on the operating system upgrade and at the same time only require fewer person-hours to maintain the system. By using this solution, we can save budget spending on upgrading system because adding two new computer labs also cost a lot.

Moreover, rooms like TV Production Studios, Distance Learning Rooms, Digital Media Labs and Networking Labs. We placed it in the same wing, which is wing B on the north-east side. It is because, this room will be connected via fiber optic cable and as we know fiber optic cable is very expensive. By doing this, we can reduce the length of cable that needed to be connected to all of this rooms and at the same time can reduce the cost of cabling.

On the other hand, we also have considers all the security measure that is needed, for example, likes CCTV which can record the activities in the building and also can be used for investigation evidence if there is theft was found. Especially for the security of the network, we consider upgrading the level of security to protect the important or private documents like student grades or student profile information. Besides that, spiral stairs are placed at the end of every wing according to our floor plan, it is used for emergency cases like the building caught on fire.

We also already identify the equipment or devices that we need to use such as access points, routers, and switches. The length of the cable that we used is sufficient throughout the building by setting the scaling of the building on our floor plan. So, it is easier to do the cabling and do not waste any cable without planning. Apart from that, we also consider the range of an access point can support and basically for now we estimate to use five access points (maybe will be changed in the future plan).

Fire Drill Floor Plan



~ Follow the red arrow to escape from the building.

Figure 8: Floorplan with emergency escape arrow

5.0 CONSIDER AND CHOOSE DEVICES

How many of each device will you need?

- a) Generator: 1
 - To support two new room for emergency case.
- b) Router: 1
 - Enough to connect to the institution network.
- c) Switches: 48 port – 2
 24 port – 3
 8 port – 4
 - Needed a total of nine switches to make a Mesh network topology.
- d) Server: 1
 - Enough to support two new rooms for server connection.
- e) Access Point: 5
 - For devices that used wireless connection or PC that have troubled with cable connection.
- f) 2-Port Faceplate: 16
 - Will be equipped in two new computer labs.
- g) 48-Ports Punchblock: 2
 - Why do we need *Punchblock*? Why not the connection *go straight to the switch*? It is generally have something to do with *ease of management*. For example, desk locations on an office floor can be cabled back to a wiring closet punchblock which is labeled with the locations. You can then connect short patches between the punchblock and the switch. This makes re-patching desk locations much simpler, as the desk -> punchblock runs don't need to be touched at all.

- For example, many server rooms have an **MDF** (Master Distribution Frame); all servers and all switch ports are cabled back to labeled patch panels on this frame. Then, creating a connection between a server and a switch is a simple case of a patch between two ports on the frame, rather than needing to have floor tiles lifted to run a new end-to-end patch.
- h) Cat6 Data Insert: 32
 - Implemented in the face plate.
- i) Cat6 Patch Cable: 2
 - To connect punchblock to the switch in Computer Labs 1 and 2 because punchblock is located near the switch.
- j) SFP Transceiver: 28
 - To be equipped in the switches for connecting the fiber optic cables to the network. Fiber optic cables need a special port for connecting to the switch, which is SFP Transceiver.
- k) Cat6 Linecord: 31
 - Used by the student PCs in the Computer Labs.
- l) Cable Cat6: 3 sets
 - Estimated length used: 2462 feet
- m) Cable Fiber Optic: 75 sets
 - Estimated length used: 2316 feet
- n) Fiber Optic Coupler:
 - To connect the same type of fiber optic connector together.

How many switches used, and how many ports in each switches?

Switches: 2 units of 48 port; 3 units of 24 port; 4 units of 8 port

Total: Nine switches used with a total of 200 ports that can be used.

What features of the each device will be a part of your decision?

Features of the each device is according to the factor:

- Cost – We chose devices that have the most reasonable price, but some of it need to be a bit expensive. This is because we do not want to pay more for the 20 years.
- Speed and types of ports – We chose the ports that can transmit data with high speed.
- Expandability – We also make sure that we reserved extra ports for future use.
- Manageability – It is important that the devices we chose are easy to manage.
- Security – We chose devices that have high security to prevent attack from the outside or malicious virus.

How we ensure network redundancy?

- Network redundancy is a process through which additional or alternate instances of network devices, equipment and communication mediums are installed within network infrastructure.
- It is a method for ensuring network availability in case of a network device or path failure and unavailability. As such, it provides a means of network failover.
- The underlying concept behind network redundancy is to provide alternate paths for data to travel along in case a cable is broken or a connector accidentally un-plugged. However, Ethernet as standard cannot have rings or loops in the network as this will cause broadcast storms and can ultimately cause the network to stop working. A certain mechanism is needed to support this type of topology.
- To achieve redundancy, the network infrastructure, which is switches, must support redundancy protocols

Terms used in the topology below:

CL: Computer Lab

TV: TV Production Studio

DLR: Distance Learning Room

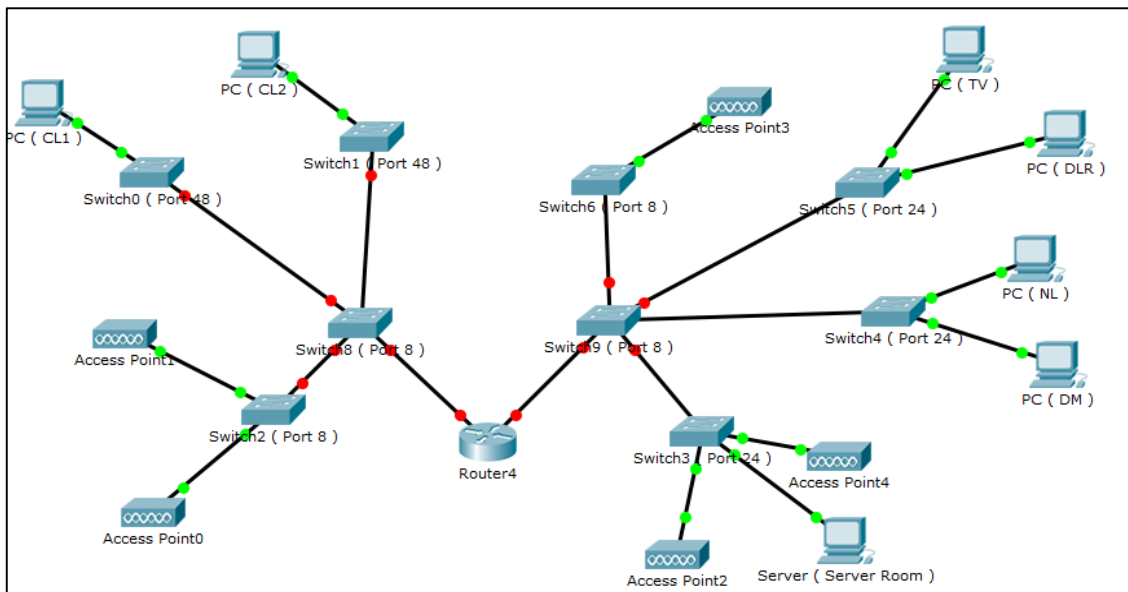
NL: Network Lab

DM: Digital Media Lab

designed to negate the usual problems of putting loops into an Ethernet network, maintaining a default data path and switching to an alternate one when a fault occurs.

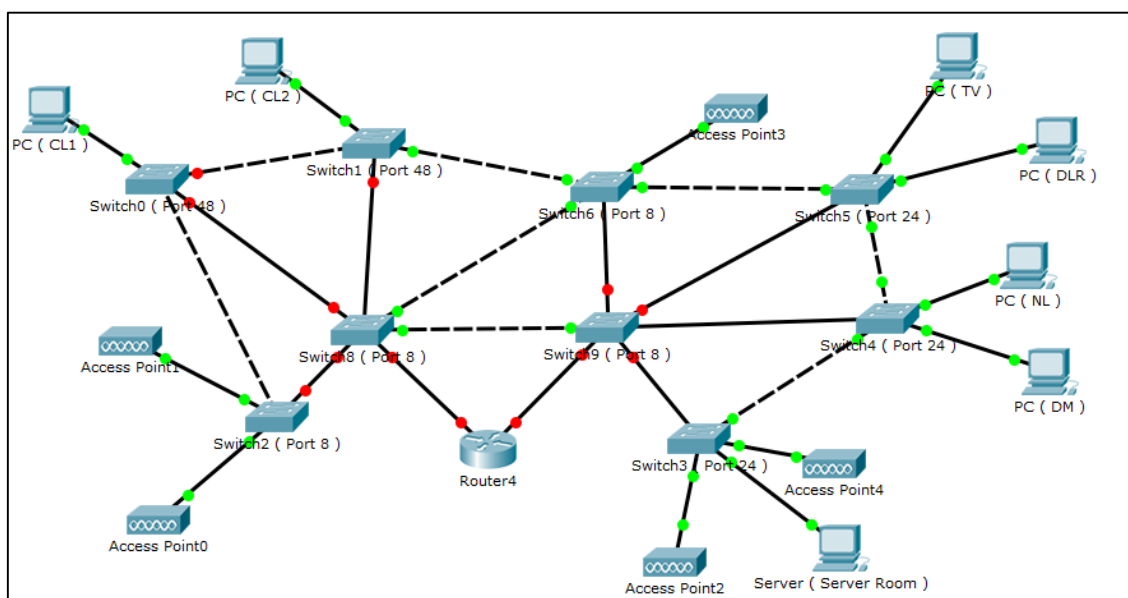
[Topology That We Design: Star Topology]

Based on the topology, we only used one router. That router is connected to two switches 8 port. After that, those switches will be connected to other switches.



[Topology That We Will Use for Network Redundancy: Mesh Topology]

To ensure network redundancy in this topology, we convert the old Star topology to Mesh topology. If one node fails, there will be other routes that can be used to send data.



6.0 REFLECTION

1) Are you surprised by the prices?

Based on the list of prices of all devices that we need for the network cabling, we are surprised by prices stated. We never expect some of the devices are quite expensive, but luckily some of it have discount. So, it can reduce our budget where we were given a budget, RM500,000.00 and we only used up RM114,674.13 to buy all the devices.

It is also not that easy to search the prices of the devices where we need to compare the prices with many different websites such as Amazon, e-bay and others. In our project, we stick to the Amazon website as it always provides promotions on some devices which have been stated in the price list table.

2) Do you have a context for this equipment and its cost?

Yes. First, the router we decided to use is an integrated service router. We decided to use this router because of it's wire-speed performance for concurrent services such as security and voice, and advanced services to multiple T1/E1/xDSL WAN rates, it can support for over 90 existing and new modules. The price per unit is RM1613.20 and the amount we bought is one only.

Then, there are three types of switch that we decided to go with which is a switch with 8 port, 24 port and 48 port. There are 4 switch with 8 port, 3 switch with 24 port, and 2 switch with 48 port. Next is for fiber optic cable. An optical fiber cable is a cable containing one or more optical fibers that are used to carry light. The optical fiber elements are typically individually coated with plastic layers and contained in a protective tube suitable for the environment where the cable will be deployed. Different types of cable are used for different applications, for example, long distance telecommunication or providing a high-speed data connection between different parts of a building. In our case, we are using it to provide a higher speed for internet connection and file transfer. The length that we estimated to use is about 2316 feet. The cost of the cable per unit is RM388.67, but fortunately we found an online store that offer a discount and decided to

buy at that store. The discount is 85% and the price after discount is RM57.59. Lastly, the sub-total price is RM20444.45.

After that is Cat6. Cat6 commonly referred to as Category 6 Ethernet Cable, is a standardized cable for Gigabit Ethernet and other network physical layers that are backward compatible with the Category 5/5e and Category 3 cable standards. Compared with Cat 5 and Cat 5e, Cat 6 features more stringent specifications for crosstalk and system noise. The cable standard provides a performance of up to 250 MHz. The length of the cable we estimated to use is about 2462 feet including the cable in case of redundancy in the future. The cost for the cable per unit (1000' per unit) is RM489.87. The sub-total of the cost for now is RM1469.61.

The conclusion is, we can conclude that all of the equipment chosen are based on the requirement. The cost of all the equipment given in Task 5 are optimized with the budget given.

3) *Have you ever considered cost as a factor for choosing networking devices?*

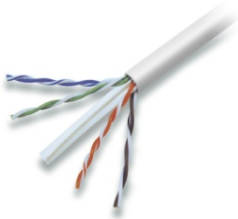
Yes, of course! We have considered all of the cost before choosing the suitable networking devices. We have chosen suitable devices that meet own requirement needed. To reduce the cost, we buy the networking devices that are included with the discount price. We want the latest and greatest technology because we don't want to waste our time on technology that will not be beneficial. We are willing to pay high cost for some devices because we don't want to pay for another upgrade for the next 20 years. Mostly, the devices that we chose is specially prepared for the future needs to the extent possible at the best value possible.

4) *What is confusing about this price list?*

When we discussing with each other about the devices list in the case study, the most confusing thing is when choosing which type of network devices that we should use for our project. Every model of that device has a different security measure, specification, and price. After surveying the advantage and price for every device, we have concluded that

these devices are the best fit for the requirement that is needed for this project. Prices and discount are referred from the Amazon.com.

[Price and Quantity List For This Project]

DEVICE	FEATURE	PRICE PER UNIT (RM)	QUANTITY	SUB- TOTAL (RM)
Cat6 1000' (Belkin# A7L704-1000-BLU) 	- It is ideal for high-end solutions such as Gigabit Ethernet, 1.2Gbps ATM, and broadband video. - This Cable consists of eight solid 24AWG wires and a separator to separate four pairs. - Plenum-rated for horizontal installations.	489.87	3 (2462')	1469.61
Generator (UPS APC SURTD3000XLI) 	- Output power capacity (VA): 3000 VA - Output power capacity (Watts): 2100 W - Emergency Power Off (EPO): Y - Nominal input voltage: 230 V - Output voltage note: 220/230/240	4518.00	1	4518.00

Access Point (Cisco AIR-AP1242AG-A-K9 Aironet 1242G) 	• Cisco Aironet 1240AG Series IEEE 802.11a/b/g access points deliver the versatility, high capacity, security, and enterprise-class features demanded by WLAN customers. • The Cisco Aironet 1240AG Series provides local as well as inline power, including support for IEEE 802.3af Power over Ethernet (PoE).	492.80	5	2464.00
Integrated Services Router (CISCO 2851) 	• Using wired. • Transfer rate: 1 gigabits per second • Ports : 2	1613.20	1	1613.20
62.5/125 MM Duplex Fiber Jumper (Belkin# f2f402LL-2m) 	• Producing the highest-quality cabling • Length: 6.56ft • First End Connector: Type-LC MultiMode • Second End Connector: Type-LC MultiMode	Original Price : 388.67 85% Off : 57.59	355 (2316')	20444.45

48-Port Cat6 Punchblock (Hubbell# P648UF19) 	• 48-port panel 110-mod 8w8p t568a/b cat6 hinged w/cbl mgm panel black 4 • Attributes Number of Ports: 48 • Wiring: A/B UNIVERSAL • Termination: 110-MOD Rack Units: 4	2296.23	2	4592.46
2-Port Data Faceplate (Hubbell# IFP12EI) 	• Provide a sleek, secure installation serving all single and double-gang multimedia applications. • Designed to flush mount with box eliminator. • Durable construction. Complemented by a broad range of voice/data/video connectors. • Provides a rear loading, snap in, flush mounting of Hubbell jacks and AV keystone connectors.	35.96	16	575.36

Data Insert Cat6 	• Confirmed contact performance over full range of (FCC Specification) deflection limits • 2,500+ mating cycles • •Insulation Displacement Contact (IDC) accommodates 22 to 26 AWG/Solid • 10+ IDC terminations for same size AWG • Accepts FCC 6-position plugs • Positive PSACR beyond 300 MHz (channel) • Exceeding IEEE 802.3ab Gigabit Specification to < 1 * 10-1	30.02	32	960.64
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5' Patch Cable Cat6 (Belkin# A3L980-05-BLU) 	• PowerSum tested • Snagless molds protect the RJ45 tap from being damaged during installation • Capable of improving the return loss in your current network by 40% • Internal load bar assures accurate positioning of pairs and consistent high performance • 2-row staggered pin configuration to control cross-talk at hyper-frequency	58.88	32	1884.16
15' Linecord Cat6 (Belkin# A3L980-15-BLU) 	• Maximize your network speed with higher bandwidth and enhance your network for future expansion • Perfect for VoIP, video, and file transfers • Supports 10/100/1000 BASE-T • Supports 10GBASE-T (up to 50 meters) • Exceeds CAT 6 performance requirements - Performance tested to exceed 500MHz • 50-micron, gold-plated connectors ensure clear transmissions	39.55	32	1265.60

	Snagless molds protect the RJ45 tabs from damage during installation			
Proline Cisco GLC-SX-MM Compatible 1000Base-SX MMF SFP (mini-GBIC) module (cisco# glc-sx-mm-cdw) 	- SFP (mini-GBIC) transceiver module 1000Base-SX - LC multi-mode - plug-in module - up to 1800 ft	402.35	28	11265.80
48-Port Switch 10/100/1000 PoE (Cisco# ws-c3560g-48ps-s) 	48 Ethernet 10/100/1000 ports and 4 SFP-based Gigabit Ethernet ports 1RU fixed-configuration, multilayer switch - Enterprise-class intelligent services delivered to the network edge - IEEE 802.3af and Cisco prestandar Power over Ethernet - IP Base software feature set (IPB)	Original Price: 23374.21 72% Off : 6693.77	2	13387.54
24-Port Switch 10/100/1000 PoE	24 Ethernet 10/100/1000 ports and 4 SFP-based Gigabit Ethernet ports 1RU fixed-configuration,	6383.38	3	19150.14

(Cisco# ws-c3560g-24ps-s) 	multilayer switch - Enterprise-class intelligent services delivered to the network edge - IEEE 802.3af and Cisco prestandard - IP Base software feature set (IPB)			
8-Port Switch 10/100 PoE (Cisco# ws-c3560-8pc-s) 	- Port Speed: 10/100 with IEEE 802.3af and Cisco prestandard Power over Ethernet (PoE) - Uplinks: 1 dual-purpose 10/100/1000 and Small Form-Factor Pluggable (SFP) port - Software Image: IP Base	Original Price: 5103.05 60.47% Off: 2011.96	4	8047.84
Cisco Syst. 2821 Security bundle w/ Adv. Sec. [Server] (CISCO2821-HSEC/K9) 	- DES,3DES, AES 128/192/256 2 10/100/1000 LAN speed - Up to 1500 VPN Tunnels	1394.03	1	1394.03
LC/LC Duplex Multimode Fiber Optic Coupler (fadp-lc-lc-mm-dx) 	- Housing construction: Plastic - Support for Duplex Multimode fiber connection - Weight: 0.010LB0	58.49	370	21641.30

TOTAL AMOUNT				114674.13
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Explanation:

We decided to add one more devices, which is a coupler for fiber optic cable called *LC/LC Duplex Multimode Fiber Optic Coupler*. This is because we want to use it to connect easily with other fiber optic cable. We using only two switches at the north-east wing, so we decided to include fiber optic coupler. It is not a bad idea after all but a proper installation and vendor consultation is needed to make sure the connection can be transmit smoothly without any problem. We managed to find the best prices for all the devices above, and of course some of them have discount included too. So, how do we estimate all the prices and quantity needed for this project? More explanation about it will be explained in Task 7 and 8.

Budget Given: RM 500,000.00

Estimated Price Needed for Upgrade in this Project: RM 114,674.13

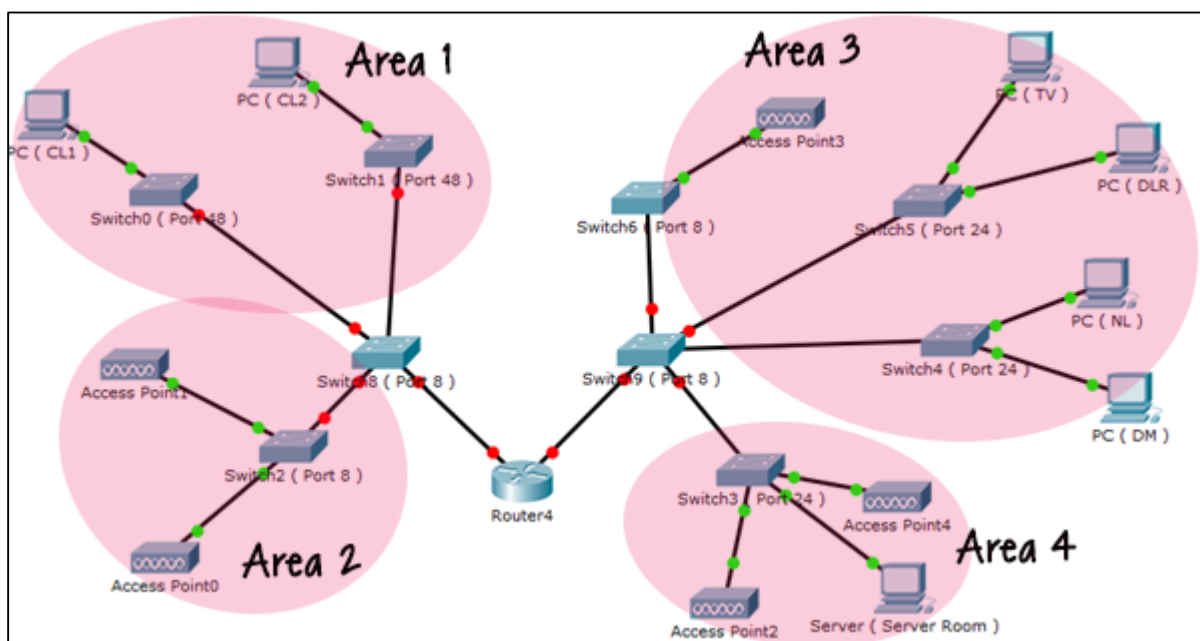
Balance = RM 500,000.00 – RM 114,674.13

= RM 385,325.87

7.0 DESCRIBE THE PHYSICAL AREAS

- *Identify the work areas on your floor plan*

To connect the devices, first we need to define each area and its needs. The work area is the space inside a building where employees, building occupants, or system users work and use their communication equipment.



Work Area 1 – The west area at wing A, which contain two computer labs which are Computer Lab 1 (CL1) and Computer Lab 2 (CL2). For CL1, switch0 (48 port) connect to the end user (student's PC) through the fifth-teen 2-port data faceplate where one 2-port data faceplate for two PC. Same method is used for CL2 that connect to switch1 (48 port).

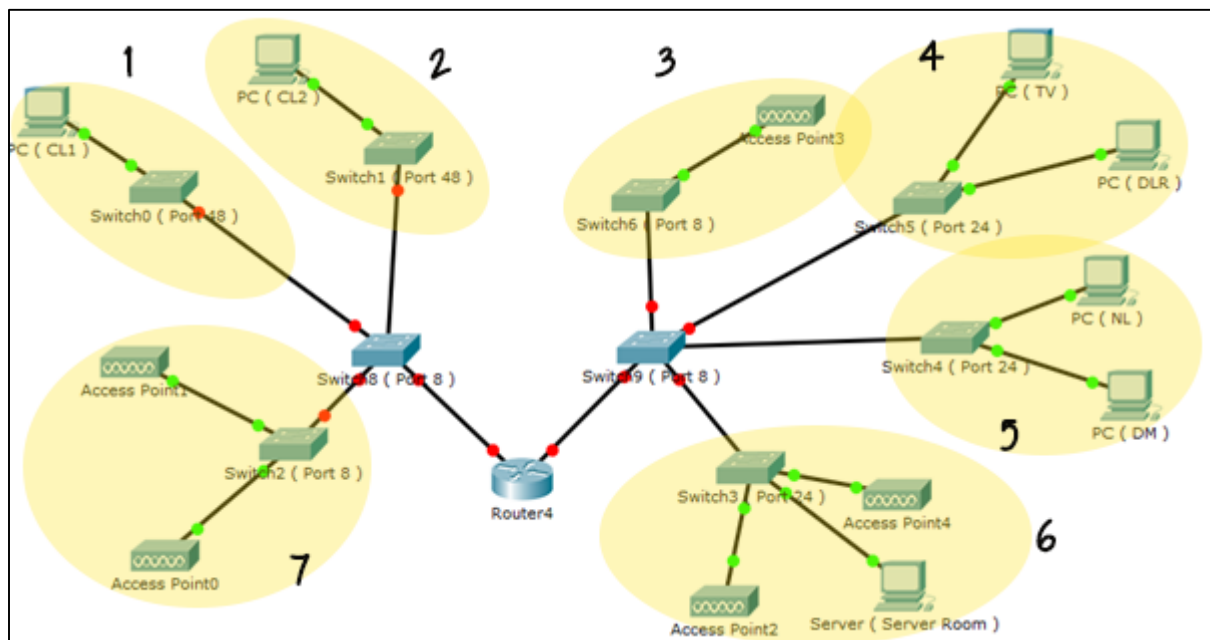
Work Area 2 – The west area at wing A, which has two Video Editing room (VD1 and VD2) and also three Video Conference room (VC1, VC2, VC3). Two Video Editing Room share an access point (access point0), meanwhile three Video Conference room share another access point (access point1). So, it's easier for end user to connect to the WiFi provided. There are four 2-port data faceplate in each Video Editing room which share switch1 (port 48) with Computer Lab2.

Work Area 3 – The north-east area at wing B, switch4 (24 port) will be connected to two Networking Labs (NL1 and NL2) and two Digital Media Lab (DM1 and DM2). Then, the

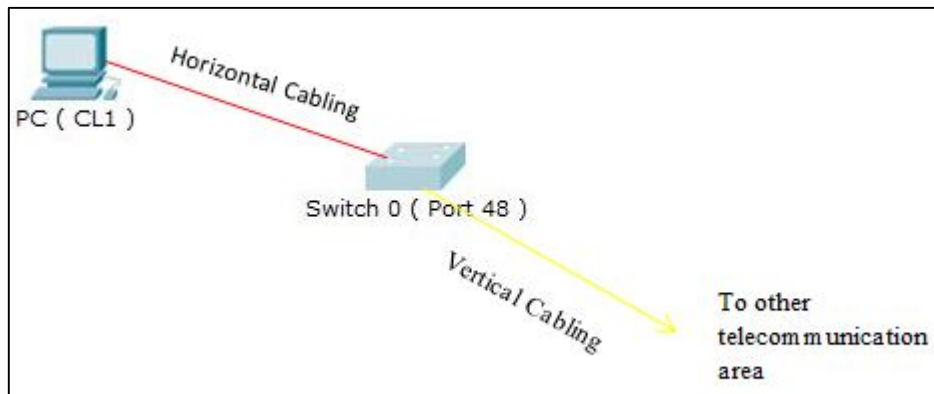
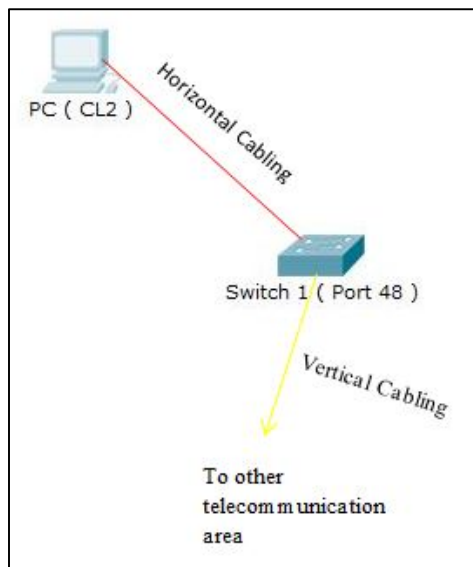
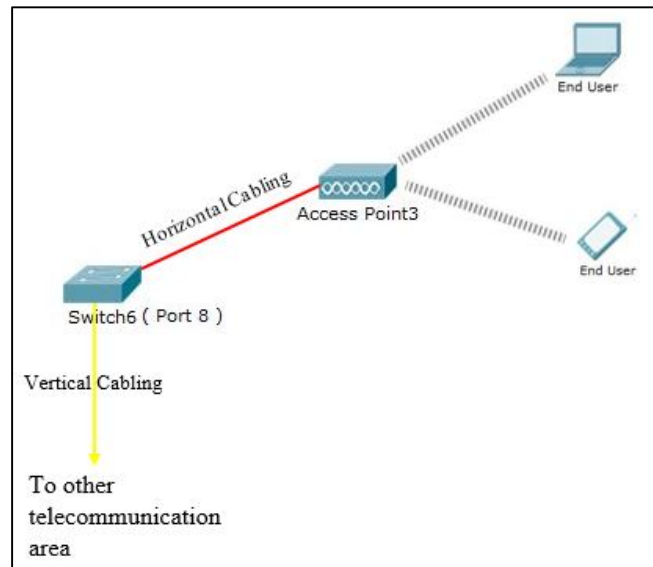
switch port will be connected to 2-port data faceplate which will connect the PC in each rooms. Besides, there are two TV Production Studio (TV1 and TV2) and two Distance Learning rooms (DLR1 and DLR2) that connected to switch5 (port 24). This switch will connected to 2-port data faceplate and then connect the PC (end user) for the internet connection. In addition for switch6 (port 8), it will connected to access point3 which will provide WiFi at meeting room, mini library and also lobby area.

Work Area 4 – The south-east area at wing C, which has a Server Room, a Technician Room, a student general lab, a restroom and a Backup Room. In this area, switch3 (24 port) will be connected to two access points, which are access point2, access point4 and a server at the Server Room. For access point2, it will be shares by the lobby of faculty (the center) and the Technician Room which allow the connection of WiFi. On top of that, for access point4, it will shared by dining area (east area), student general lab (SGL) and restroom (RR).

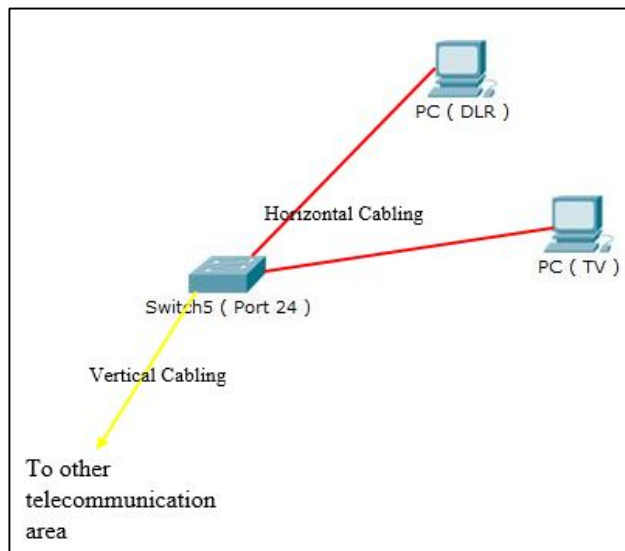
- *How many connections have you determined you need?*



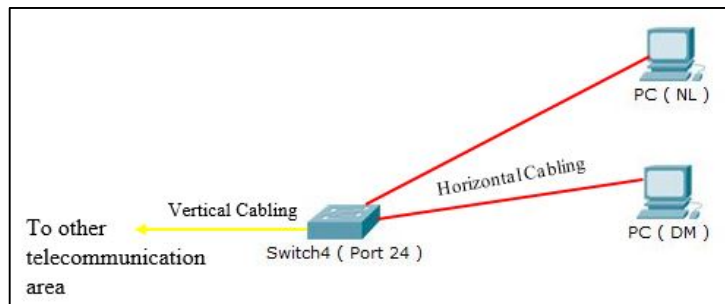
Based from the topology above, we determined that we need seven (7) connections for this project.

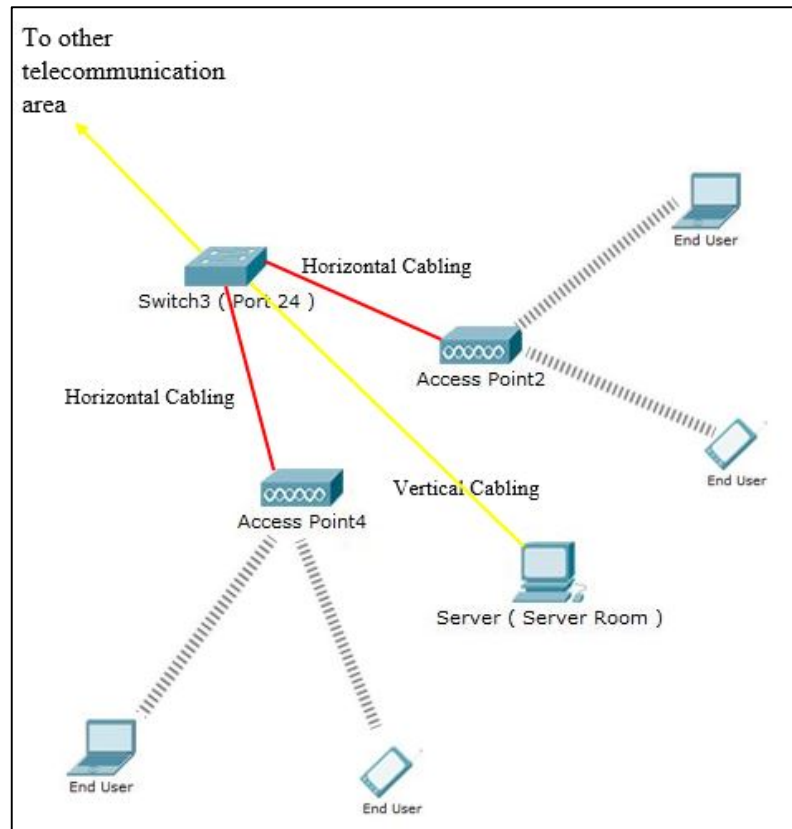
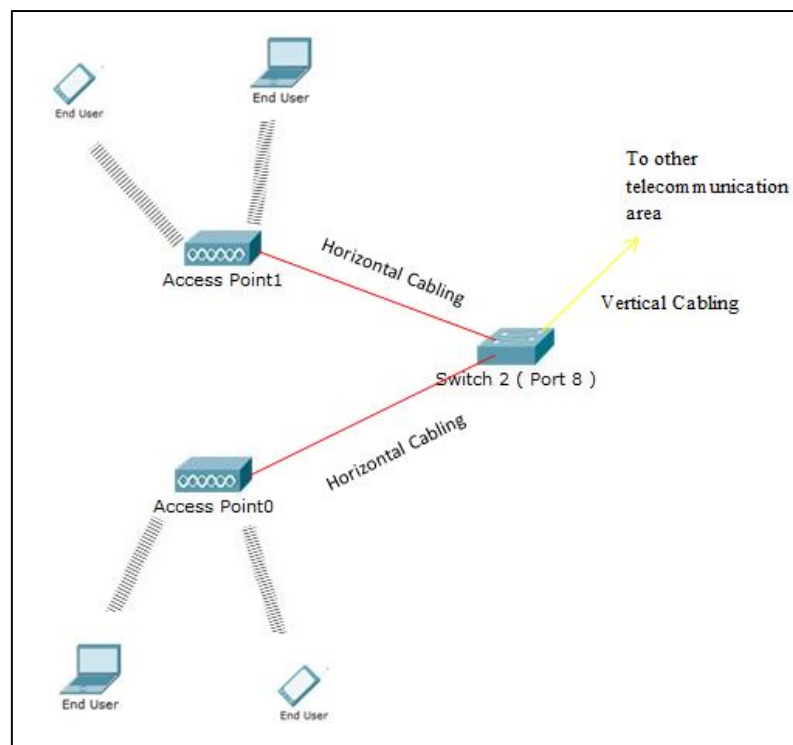
Connection 1**Connection 2****Connection 3**

Connection 4



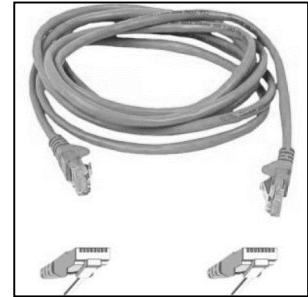
Connection 5



Connection 6**Connection 7**

- ***How many patch cords will you need?***

For the patch cord (patch cable), we use 5' Patch Cable Cat6 (Belkin# A3L980-05-BLU). According to what we have found, price for a unit of this cable is about RM58.88. We will use about 32 patch cords because we consider that this short cables is use for connecting the patchblock to the switch in Computer Lab 1 and Computer Lab 2. So, the total price now is RM1884.16.



- ***How many switch ports?***

For this project, we planned to use two 48-port switch, three 24-port switch and four 8-port switch. A total of nine switches is needed to make a Mesh network topology. The 48-port switches are used at Computer Lab 1 and Computer Lab 2. The 24-port switches are used at northeast wing of the building. Lastly, the 8-port switches are used for connect all five access point to the network.

Total price needed to pay for all the switches is about RM240,068.20. Besides, we also chose switches that have **PoE**. Power over Ethernet (PoE) is a technology for wired Ethernet LANs (Local Area Networks) that allows the electrical current necessary for the operation of each device to be carried by the data cables rather than by power cords. Doing so minimizes the number of wires that must be strung in order to install the network. The switches that we pick are able to transmit data with high speed accordingly to what the key personnel in the School of Telecommunication want.

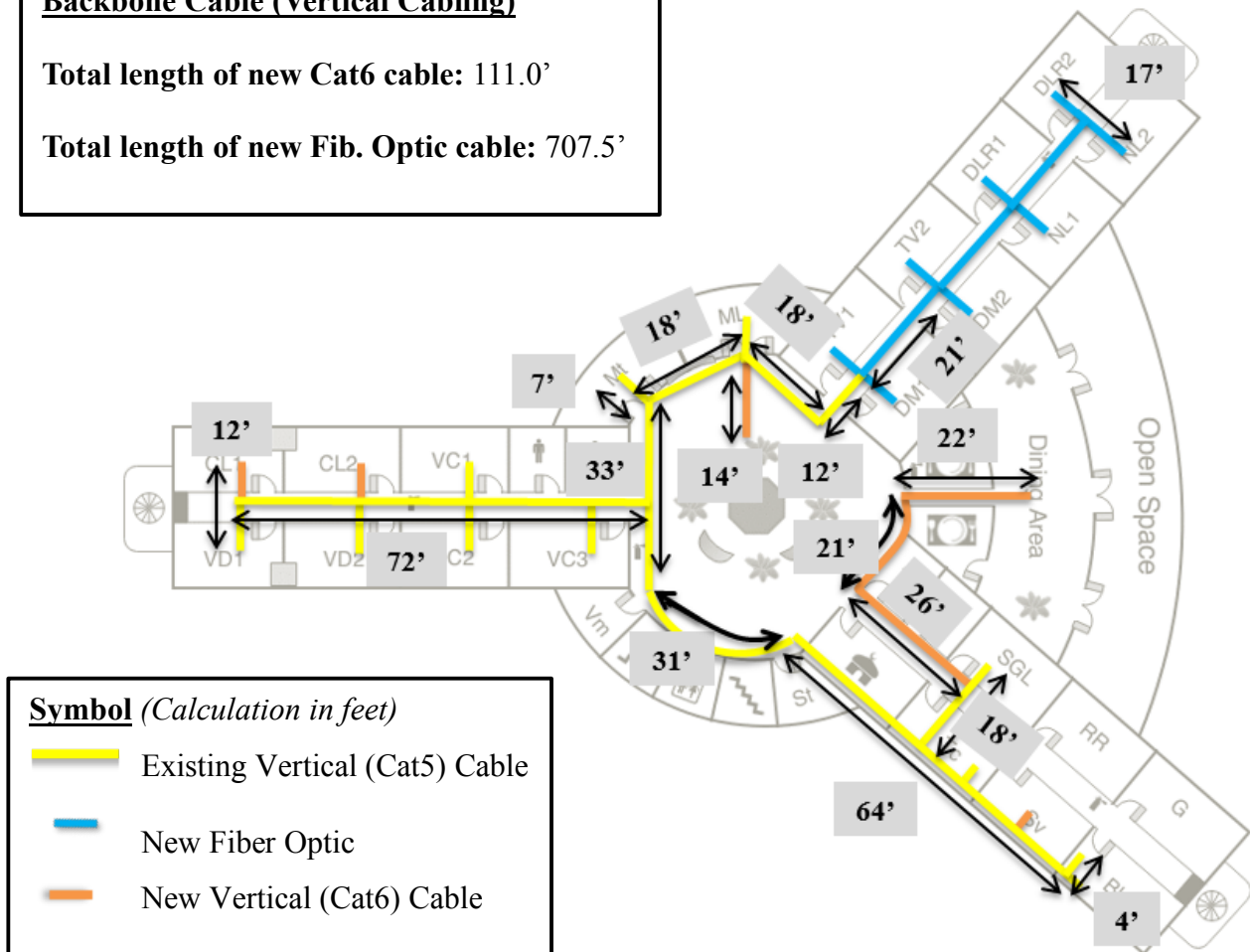
8.0 TYPES OF MEDIA & HOW MEDIA CONNECT

▪ *Vertical Cabling*

Backbone Cable (Vertical Cabling)

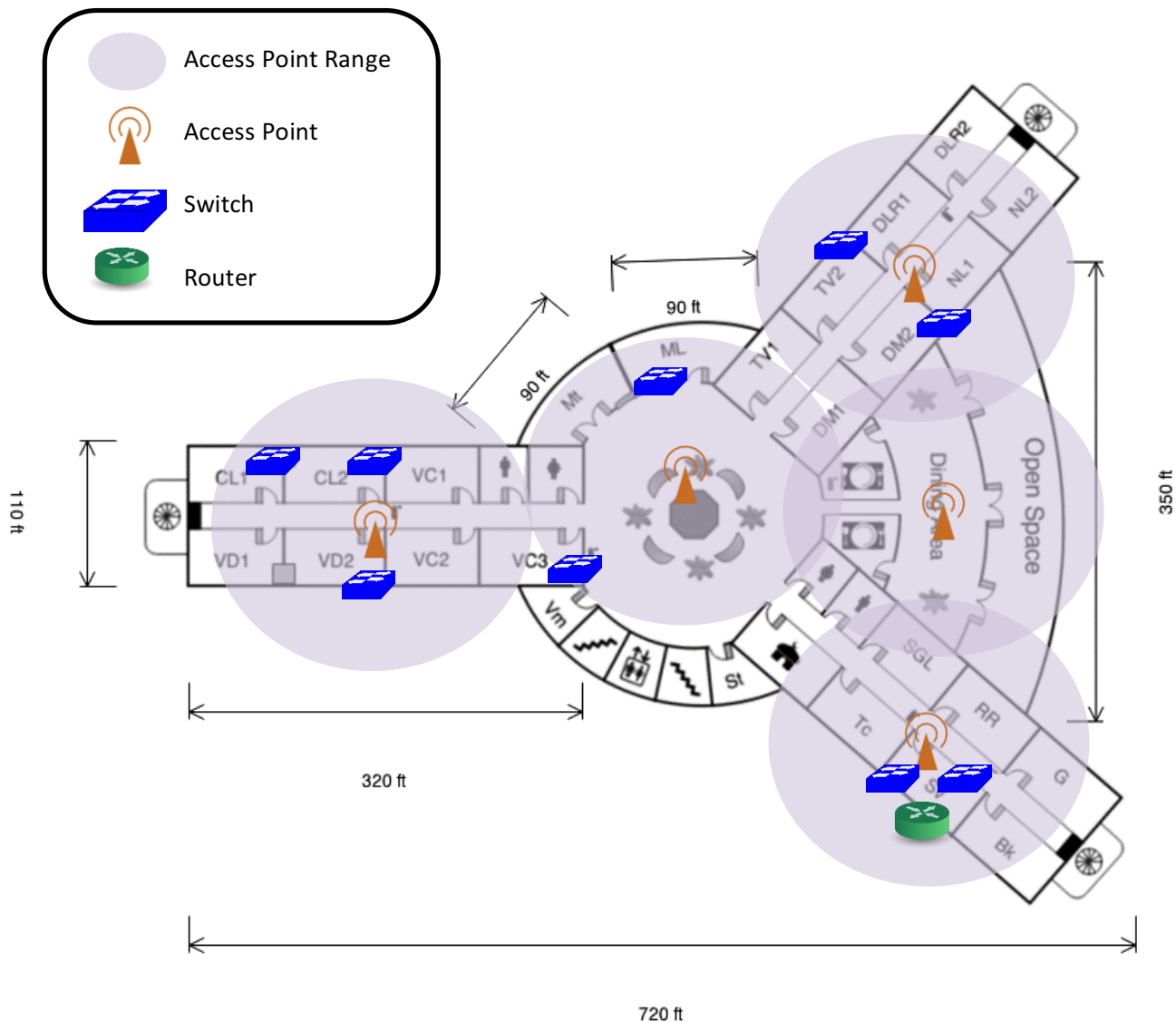
Total length of new Cat6 cable: 111.0'

Total length of new Fib. Optic cable: 707.5'



According to the case study, we already have existing Vertical (Backbone) cable which is Cat5 cable. To satisfied the needs of what the key personnel of this institution want, for the new Vertical cable to connect the new and upgraded rooms, we will be using Cat6 cable and at the north-east side by using fiber optic cable. Extra cables are needed at the Server Room for the new Computer Labs to connect to the institution server and the internet, and also at the Dining Area and Mini Labrary (ML) to connect to a new Access Point. All price calculation for both type of cable will be shown at **page 49**.

▪ *Access points, Router and Switches Location and Range*



All the devices location above are based on the topology that we made in Task 7. There is a total of nine (9) switches, one (1) router and five (5) access point in the diagram. The range of the access point (*up to 150' (46 m) to 300' (92 m)*) is also included.

▪ *Horizontal Cabling in Upgraded Rooms*

Lots of faceplates will be implemented on the floor instead on the wall. Detail of all estimation and explanation will be given below. All calculation are in **inch** and **cable price calculation will be shown on page 49**. Assume that the length needed for the cable to plug in to the PC and to the punchblock is about **15.7 inch** and the length from the switch to the wiring area which is near the ceiling is about **80.0 inch**. Wall height is **96 inch**. Cable length needed to reach the wiring area for span to another room is about **95.7 inch**. (All estimation done based from the picture below).

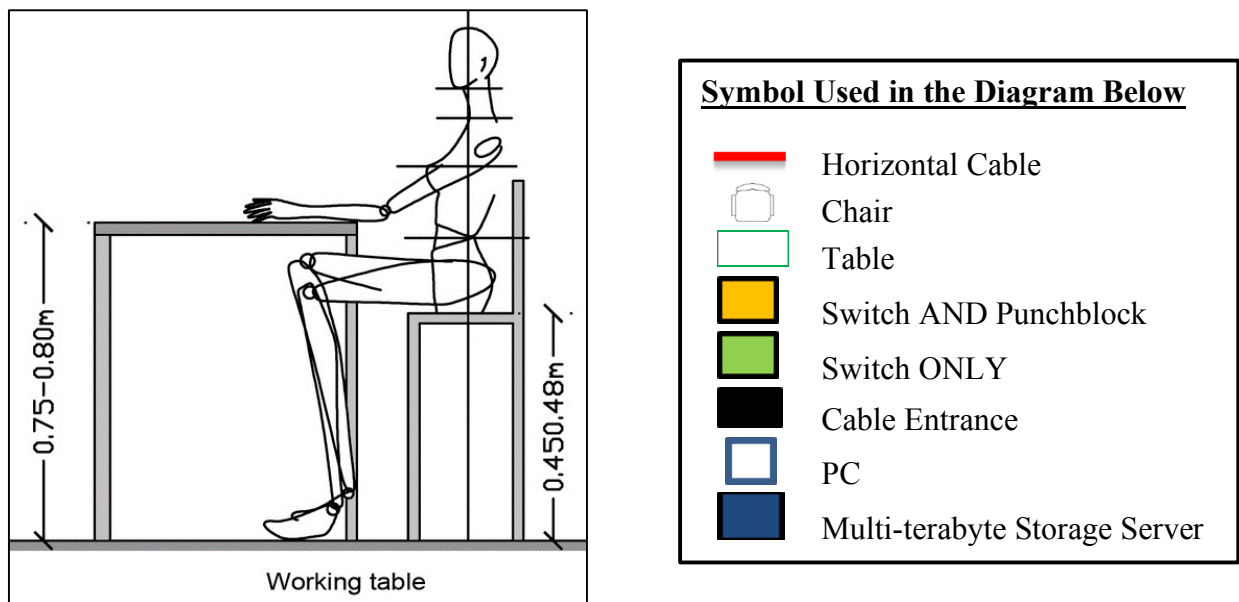


Figure 9: Working Table

(http://www.decosoup.com/images/stories/Design_Issues/Body_measurements/table_height.jpg)

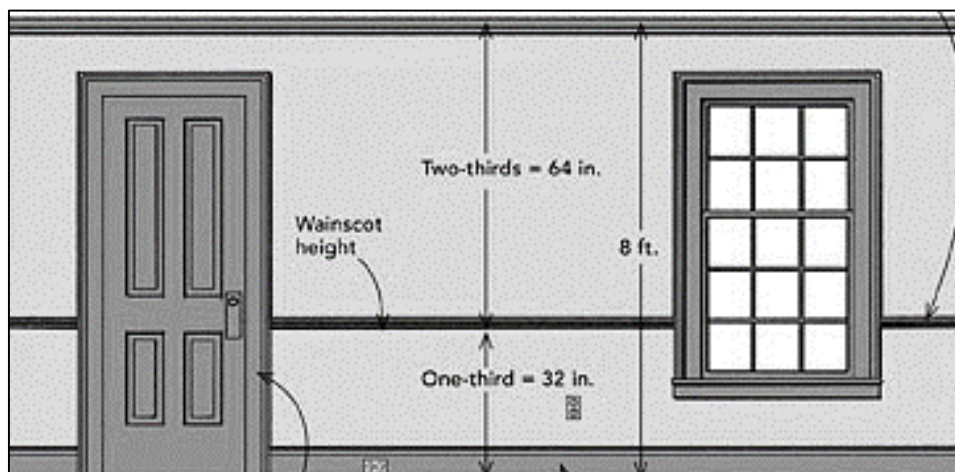
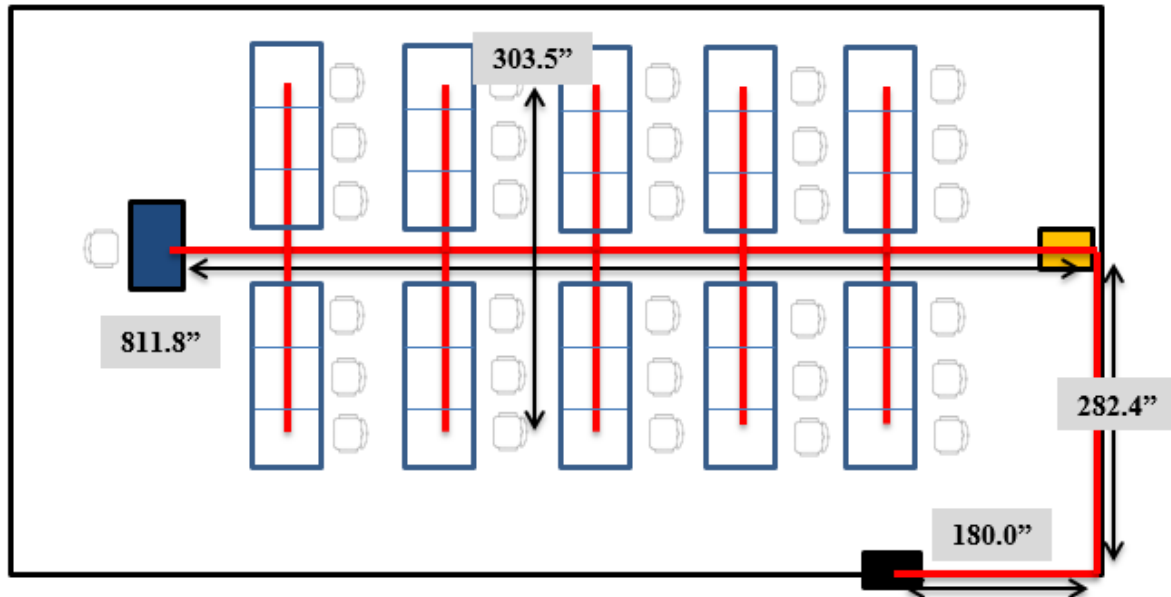


Figure 10: Typical wall and door height

(http://www.finehomebuilding.com/CMS/uploadedimages/Images/Homebuilding/Departments/021193db124-01_lg.jpg)

1) Computer Lab 1 / Computer Lab 2 (CL 1 / CL 2)



Explanation: There will be one 48-port switch and one 48-port punchblock in each CL 1 and CL 2. There are 30 PC and 1 multi-terabyte server. All the faceplates will be equipped on the floor. The amount of faceplate that will be used is about 16 units. Type of cable used here is Cat6 cable.

Computer Lab 1 / Computer Lab 2

Total Floor-to-PC Cable length: $15.7'' \times 31 = 486.7''$

Total Floor-to-Switch Cable length: $15.7'' \times 32 = 502.4''$

Switch-to-Wiring Area Cable length: $80.0''$

Total Cable Length on the Floor: $12674.8''$ (32 units of cables)

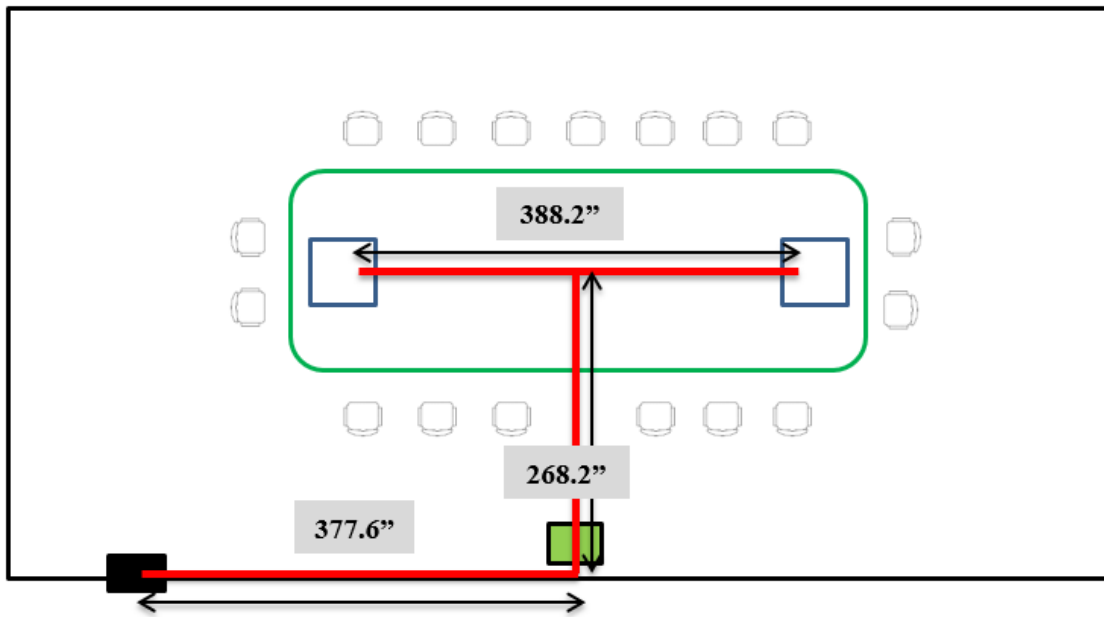
Total length of Cat6 cable: $12674.8'' + 486.7'' + 502.4'' + 80.0'' = 13743.9''$

Total Length for Both Rooms: $27487.8''$

Faceplate: $RM35.96 \times 16 = RM575.36$ (16 units / 32 ports)

Punchblock: $RM2296.23$ (1 unit)

2) TV Production Studio 1 / TV Production Studio 2 (TV 1 / TV 2)



Explanation: There will be one 24-port switch in TV 2 only. There are two PC with a large table in the middle. Type of cable used here is Fiber Optic cable (*Belkin#f2f402LL-2m*).

Note: ONLY one switch is added, which is located in TV 2. Everything else in TV 2 is same for TV 1. **Distance Learning Room 1 and 2** will be sharing the same switch in TV 2.

TV Production Studio 2

Total Floor-to-PC Cable length: $15.7'' \times 2 = 31.4''$

Total Floor-to-Switch Cable length: $15.7'' \times 2 = 31.4''$

Total Cable Length on the Floor: 924.6'' (2 units of cables)

Total length of Fiber Optic Cable: $924.6'' + 31.4'' + 31.4'' = \underline{1161.1''}$

Total Switch-to-Wiring Area Cable length: 80.0''

Length of Cat6 Cable: $377.6'' + 80'' = \underline{457.6''}$

TV Production Studio 1

Total Floor-to-PC Cable length: $15.7'' \times 2 = 31.4''$

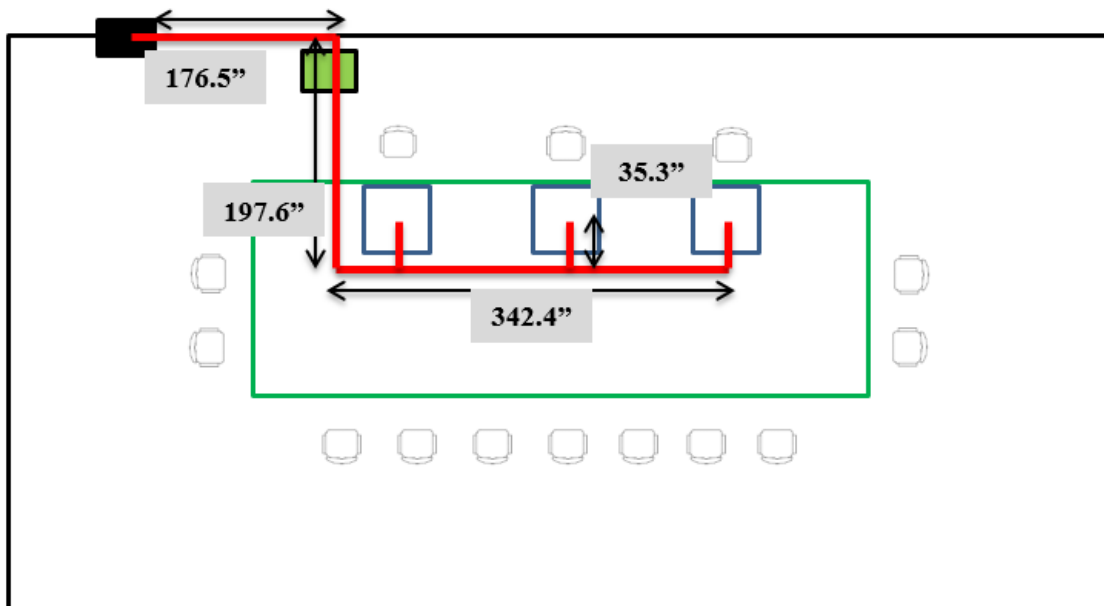
Total Floor-to-Ceiling Cable length: $95.7'' \times 2 = 191.4''$

Total Cable Length at Wiring Area: $377.6'' \times 2 = 755.2''$

Total Cable Length on the Floor: 924.6'' (2 units of cables)

Total length of Fiber Optic cable: $924.6'' + 31.4'' + 191.4 + 755.2'' = \underline{1902.6''}$

3) Digital Media Lab 1/ Digital Media Lab 2 (DM 1 / DM 2)



Explanation: There will be one switch added in DM 2. There are three PC with a large table in the middle. Type of cable used here is Fiber Optic cable (*Belkin# f2f402LL-2m*).

Note: ONLY one switch is added, which is located in DM 2. Everything else in DM 2 is same for DM 1. **Network Lab 1 and 2 (NL 1 & 2)** will be sharing the same switch in DM 2.

Digital Media Lab 2

Total Floor-to-PC Cable length: $15.7'' * 3 = 47.1''$

Total Floor-to-Switch Cable length: $15.7'' * 3 = 47.1''$

Total Cable Length on the Floor: 1298.8'' (3 units of cables)

Total length of Fiber Optic Cable: $1298.8'' + 47.1'' + 47.1'' = \underline{1393.0''}$

Total Switch-to-Wiring Area Cable length: 80.0''

Length of Cat6 Cable: $176.5'' + 80'' = \underline{256.5''}$

Digital Media Lab 1

Total Floor-to-PC Cable length: $15.7'' * 3 = 47.1''$

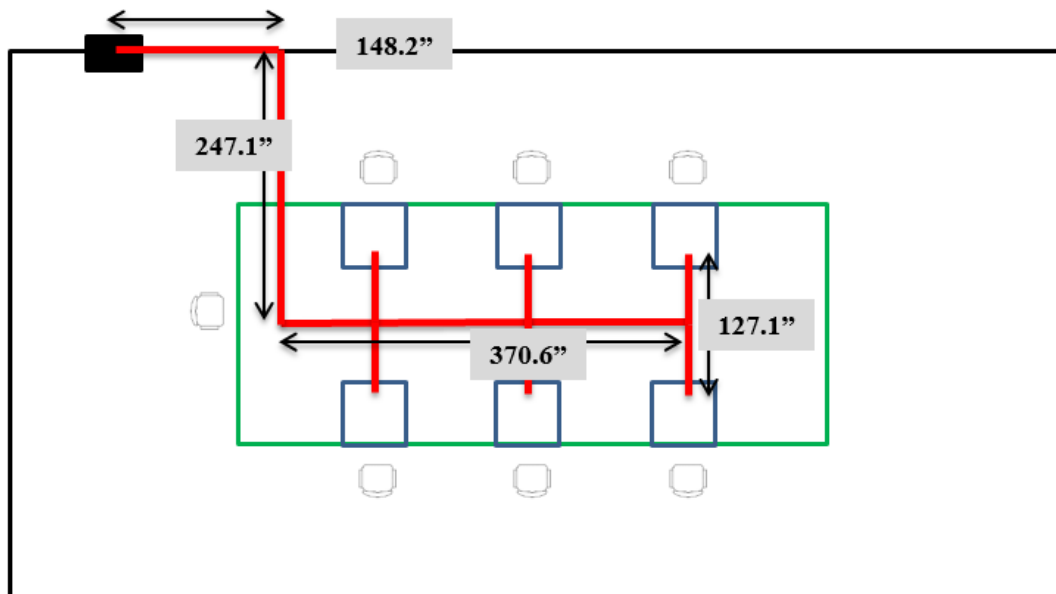
Total Floor-to-Ceiling Cable length: $95.7'' * 3 = 287.1''$

Total Cable Length at Wiring Area: $176.5'' * 3 = 529.5''$

Total Cable Length on the Floor: 1298.8'' (3 units of cables)

Total length of Fiber Optic cable: $47.1'' + 287.1'' + 529.5'' + 1298.8'' = \underline{2162.5''}$

4) Network Lab 1 / Network Lab 2 (NL 1 / NL 2)



Explanation: There are six PC with a large table in the middle. Type of cable used here is Fiber Optic cable (*Belkin# f2f402LL-2m*). Both rooms have the same structure and cable length.

Network Lab 1 / Network Lab 2

Total Floor-to-PC Cable length: $15.7'' \times 6 = 94.2''$

Total Floor-to-Ceiling Cable length: $95.7'' \times 6 = 574.2''$

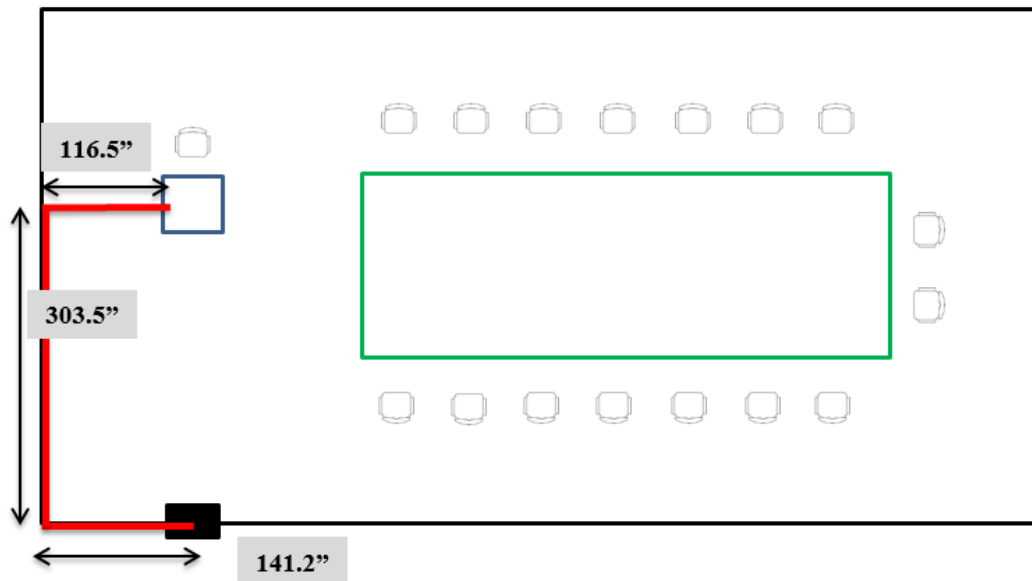
Total Cable Length at Wiring Area: $148.2'' \times 6 = 889.2''$

Total Cable Length on the Floor: 4108.5'' (6 units of cables)

Total length of Fiber Optic cable: $4108.5'' + 94.2'' + 574.2'' + 889.2'' = 5666.1''$

Total Length for Both Room: $5666.1'' \times 2 = \underline{11332.2''}$

5) Distance Learning Room 1 / Distance Learning Room 2 (DLR 1 / DLR 2)



Explanation: There are one PC with a large table in the right middle Type of cable used here is Fiber Optic cable (*Belkin# f2f402LL-2m*). Both rooms have the same structure and cable length.

Distance Learning Room 1 / Distance Learning Room 2

Total Floor-to-PC Cable length: 15.7"

Total Floor-to-Ceiling Cable length: 95.7"

Total Cable Length at Wiring Area: 141.2"

Total Cable Length on the Floor: 420.0" (1 units of cables)

Total length of Fiber Optic cable: $= 15.7'' + 95.7'' + 141.2'' + 420.0'' = 672.6''$

Total Length for Both Room: $672.6'' * 2 = \underline{1345.2''}$

▪ ***Cable Length Calculation***

Cat6 Cable

$$\text{Total} = \text{Main} + (\text{CL 1} + \text{CL 2}) + \text{TV 2} + \text{DM 2}$$

$$= 111.0' + 27487.8'' + 457.6'' + 265.5'' \quad (\text{Convert to feet / '})$$

$$= 111.0' + 2290.65' + 38.13' + 22.125'$$

$$= 2461.9' = \underline{\underline{2462'}}$$

$$\text{Cat6 1000' Price} = \text{RM489.87 per unit (1000 feet per unit)}$$

$$= \text{RM489.87} \times 3 \text{ units} = \underline{\underline{\text{RM1469.61}}}$$

Fiber Optic Cable

$$\text{Total} = \text{Main} + (\text{TV 1} + \text{TV 2}) + (\text{DM 1} + \text{DM 2}) + (\text{NL 1} + \text{NL 2}) + (\text{DLR 1} + \text{DLR 2})$$

$$= 707.5' + 3063.7'' + 3555.5'' + 11332.2'' + 1345.2'' \quad (\text{Convert to feet / '})$$

$$= 707.5' + 255.3' + 296.3' + 944.4' + 112.1'$$

$$= 2315.6' = \underline{\underline{2316'}}$$

$$\text{Fiber Optic Price} = \text{RM57.59 per unit (6.56 feet per unit)}$$

$$= \text{RM57.59} \times 355 \text{ units} = \underline{\underline{\text{RM20444.45}}}$$

▪ ***Factors That Need to Be Consider***

1) Does the cable need to span across a room or from building to building?

Yes. Based on our planning for this project, we used different type of cables for different purposes. So, some of the cables need to span across a room for connection. The reason to do this is to make the installation of cable easier and do not wasting time, cost and the length of cable. Besides that, we using this method to make sure that the cables are connected neatly and efficiently.

In our project, we are using two types of cable which are CAT6 and Fiber Optic. The total length of CAT6 been used in this project is **2462** feet while total length of Fiber Optic is **2316** feet. The process of span cable across a room can be proved by using images provided in this Task 8 (As shown before).

2) Cost - Does the budget allow for using a more expensive media type?

Yes. In this project, our group still has balance from the budget given. From the start, we are given RM500,000 to complete this task and by this amount, it is impossible that we could not complete this task. After we made a price list for the media type or equipment that we need, overall from the price list that we made, it's only cost **RM114,674.13** for 16 type of devices and equipments. As the conclusion from the table of price list, the balance left is **RM385, 325.87** and it is still a big amount.

Based on balance that we have, we still available to buy more expensive media type or other equipments for other purposes to upgrade any parts of the building that need upgrading. It is possible to do that because the other equipment that we bought also had provides discount. Some of the equipment provide great discount and from that we can save budget.

When buying the equipment, we think wisely and decide which equipment that we need to avoid waste of money. Besides that, we also survey the price before decide which item that we want to purchase.

As conclusion, it's doesn't matter if we want to purchase more expensive of media type as it's does not effect at all our budget given which is RM500,000.

3) Bandwidth - Does the technology used with the media provide adequate bandwidth?

Yes. The technology that we used for this project and the media that we bought will provide adequate bandwidth.

Based on the arrangement of access points and switch ports that we set in this task, it will cover the area that need connection so that it can connect to the internet. In this project, we decided to use five access points which located in different areas to provide Wi-Fi for adequate bandwidth.

In our group planning for this task, the devices in a network have different bandwidth requirements. For example, a server generally has a need for more bandwidth than a computer dedicated to a single user. Hence, in this project, switch3 (port 24) is used in the area where the server room located.

Currently, the technology used in fiber-optic media like in our floor plan, wing B that will fiber optic for cabling, offers the greatest bandwidth available among the choices for LAN media. Given the seemingly unlimited bandwidth available in fiber cables, much greater speeds for LANs are expected.

4) Do we have the ability to install the cable or do we have to hire a vendor?

Based on the cable we are going to use for the project given, we have the ability to install the regular cable but for the fiber optic cable, we decided to hire a vendor since we want to make sure that all the cabling are secure and installed properly.

We will provide the remuneration and supply list for the vendors and they will know what is the right equipment they will use. There will be some phase that the vendor will go through when installing the cables such as the planning phase, drawing of schematics and plans, installing and testing phase and finishing phase.

Firstly during the planning phase, we will plan the arrangement of all the wire including the fiber optic. This is to make sure that the final position of the wires does not obstruct people in the building. For the fiber optic cable, vendors will need to ensure that there is fiber optic service available in the area from companies that provide fiber optic connection. If so, a distribution box should be located somewhere in the area. They will also locate the nearest box and run fiber optic cable from the box to the building.

Then, we will draw the schematics and plans for the cabling because documentation of the design will help us and the vendors to identify any potential problems before starting the installation.

After that, during the installation and testing phase, we will work together with the vendors because we likely need a professional on hand to carry out the plan and to provide the necessary hands-on-expertise. Furthermore, each length of cable should be a few inches longer than the space requires, allowing room for unforeseen obstacles. There will also be some slack so that the cable will not be overstretched. Then, we will test each section to make sure that the cabling done is flawless.

Finally, once all of the cables are run through the appropriate spaces and all of the closures and wall jacks are in place, the computers and other devices can be hooked up and turned on and used.

5) *Susceptible to EMI/RFI - Is the local environment going to interfere with the signal?*

The interference is actually difficult to avoid, especially in urban areas where the wireless revolution is well underway.

Thus, there are going to be some area that will experience the interference especially that are close to a device that transmit electromagnetic wave. You can't sweep away the interference problem. Elevators, microwave ovens, cordless phones, Bluetooth devices, wireless video cameras, outdoor microwave links, wireless controllers, fluorescent lights, WiMAX devices, speaker system and even bad electrical connections are some examples of a device that can cause a connection disturbance.

For prevention, the source of the disturbance should be moved away from the affected equipment or the equipment should be placed somewhere distant from the source of offending EMI. If those measures fail or are impractical, the source of EMI should be isolated from the surrounding environment. This can be accomplished by properly grounding the machinery, constructing a Faraday cage around it, or placing it in a room constructed of conductive material. The picture below is the example of a Faraday cage.



Figure 11: Faraday cage.
(<http://www.herzan.com/sites/default/files/Faraday-Cage-transparent.png>)

9.0 COMPILE AND MAKE A REPORT

“Produce a group report with all the revised and corrected steps and tasks.”

- Many changes have been made to this report.
- Format used: 12px Font, 16px Title, Times New Roman, Justify.
- Table Content, Dedication and Acknowledgement, and References are added.
- Header and Footer is included too. Report looks more formal.
- Page number is located at the bottom right. Cover is page 0.

10.0 MAKE A PERSONAL REPORT

“Write a personal report. This is individual work. Report should be authentic and unique.”

- In order: Radhi -> Tiffany -> Syakir -> Natalie
- All of the personal reports have their own unique cover and style.
- Total of four personal reports are included in this task.
- Maximum of 2 pages and have the same format as mentioned earlier in Task 9.



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FACULTY OF COMPUTING

SEMESTER II - 2014/2015

SCSR1213 – NETWORK COMMUNICATIONS

PERSONAL REPORT – CASE STUDY

MOHAMAD FAKHRULRADHI BIN MOHD SHA'RI

A14CS0167

950530-06-5311

SECTION 08 - 1SCSR

Group Name: *Fantastic 4*

Group Number: 05

Contribution in Group:

- Group Representative.
- Report final checking.
- Repaired the logic of diagram in Task 8.
- Spelling and grammar checker.
- Report formatting and compiling.
- Gave some explanation about the project to other team members.
- Did the network redundancy explanation.
- Divide the work with everyone.
- Did the references.

Group Members Contribution:

Syakir Zufayri Bin Roslan

- Estimate the length of cable in Task 6.
- Our floorplan designer, including measurement labeling in it.
- Repaired the floorplan logically.
- Helping Natalie in making the diagram.
- Did many explanations of devices context at Task 6.
- Made connection diagram in Task 7 with Natalie.

Tiffany Anak Garan:

- Done the price list table in Task 6.
- Compare price for all devices needed in Task 5 – 6.
- Did a lot of essay to answered project's questions.
- Floorplan's room labeling.
- Providing source for improving our project report from the senior.

- Did the 'Fire Drill floorplan' diagram.
- Calculated budget needed in this project.

Natalie Nuai Anak Gerald:

- Did the explanation for the floorplan.
 - Compare price for all devices needed in Task 5 – 6.
 - Made all the diagrams in Task 8.
 - Floorplan's room labeling.
 - Topology designer in task 5 - 7.
 - Helping me in adding the arrows and labels in diagrams Task 8.
- Everyone love to share ideas and help each other when making this report. Especially when making floorplan and the topology for the first time.

How I Work As a Group:

- Helped team member as much as I can.
- Making a WhatsApp group to alert and interact with other members.
- I try to make sure that the works equally divided to everyone and they like to do them.
- Provide them some tips and source for their works.
- Encourage each other to form good partnerships.
- Contribute as much as I can to the team.

Comments on the Project:

“One of the toughest and longest report that we have done in our life. I gain many new experience. Making a report about floorplan, cabling and measurement estimation had really taught me many things such as thinking logically, know how to differentiate between router, modem and, switch, and calculating budget for a report. When making this project, I got to know more about what network is all about. We need to search all kinds of things and new term (such as PoE and MDF) just to make sure the fact we stated is right and reasonable. As a Computer Network student, I'm really happy with this report. Thank you Syakir, Tiffanny and Natalie because be in my team. “



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FACULTY OF COMPUTING

SCSR1213

Network Communication

Name : Tiffanny Anak Garan

Matrix Number : A14CS0182

NRIC : 950916-13-6282

Year programme : 1 SCSV

Section : 08

Group Name : Fantastic 4

Lecturer Name : Encik Muhalim

My name is *Tiffany Anak Garan* which from the group of *Fantastic 4*. Finally, our group has done all the tasks given which is from task 1 until task 9. In this project, me as one of the member of *Fantastic 4* has contribute a lots and same goes to my other group members.

One of my contributions in this group is searching the price list for all equipments that we need for the new upgrade of the building, adding two new computer labs and also cabling. While searching the prices for each item that we planned, I consider the prices of items by compare the prices and choose the best prices and qualities according to our needs in this project. Some of the items had been given discounts, so it's help a lots in our group budget planning. It is because the budget given to our group only RM500K but at last we successfully spent it wisely and still have balance which is RM385,325.87. Besides that, I also contribute through other parts like made an explanation for questions been asking in this project.

My other group members also have done their best in completing all tasks given in this project. *Mohamad Fakhrulradhi Bin Mohd Sha'ri* as our group leader in this project had guide us wisely throughout this project until it could be done completely. He also contribute a lots in this project where he searching for the important information, explaining to his group member of the term used or information to his group member and he also made a explanation for the questions given according to the needs from each tasks.

Next, *Syakir Zufayri Bin Roslan* who contributes in the design of our floor plan after combined all of our ideas. He quite creative can draw out floor plan that can make all group members satisfy with it. Besides that, he also gave a lot of ideas in this project and also made an explanation for questions given in this project. Apart from that, he also give consideration and cooperation while doing all the tasks through this project.

Natalie Nuai Anak Gerald is also one of our group members. She is someone that do work efficiently and fast in completing the tasks given. Throughout this project, she also contributes a lot where she made an explanations and giving examples according to our project requirements. She also contribute in calculating the length of cables and draw it out so that easy to refer for future.

Throughout this project, the four of us work as a team to make sure all tasks can be completed before the due date given and all the content been arranged neatly. For every task,

we will having small discussion to contribute our ideas and compiled it as one output. After that, our group leader, *Mohamad Fakhru radhi Bin Mohd Sha'ri* will divide works to four of us according to what we want to do. So, we as a team always support each other when we lacks some parts and make sure it can be completed perfectly. Sometimes, we spent our times to complete some task together when it's need four of us to be there to settle any obstacles that we meet.

In conclusion, after completing all task in this project, I learned a lots about what is Network Communication subject and through the hardship and obstacle that me and my group member faced. This project really taught me and my group member about the process of cabling and how to plan the structure of the new upgrade of building look like. I hope I can make a project like this as a team in future because it's help a lots in our study to understand the theory that we learned from class.



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SCSR1213 – NETWORK COMMUNICATIONS

TASK 10

(PERSONAL/INDIVIDUAL REPORT)

GROUP NAME:

Fantastic 4

NAME: SYAKIR ZUFAYRI BIN ROSLAN

MATRIC NO: A14CS0139

LECTURER: EN. MUHALIM BIN MOHAMED AMIN

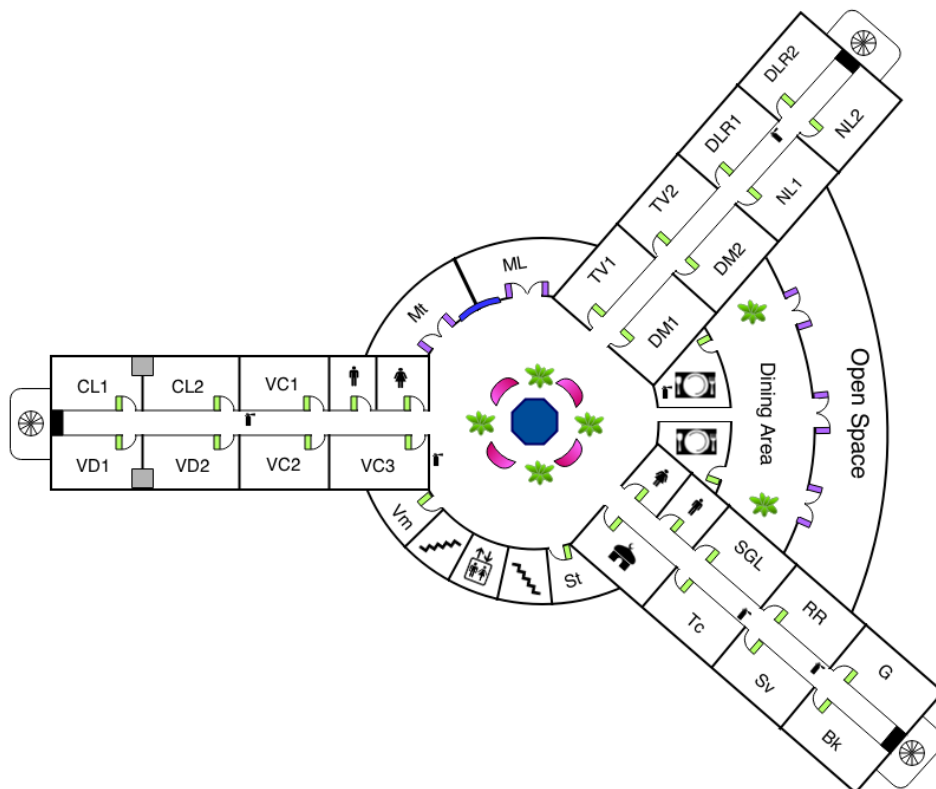
GROUP NAME

First of all, Assalamualaikum. Our group's name is *Fantastic 4*. Actually we were very confused on what our group is going to be called but then we decided for it to be *Fantastic 4* because our group is consist of 4 people which is me, Syakir Zufayri bin Roslan, Mohamad Fakhrulradhi bin Mohd Sha'ri, Tiffanny Anak Garan, Natalie Nuai Anak Gerald.

CONTRIBUTION

One of my contribution to the group is that for the plan floor, I drew it using an online application called *draw.io*. Initially, we decided to make floor plan that is shaped like an X but then when we think about it, we decided to go with the current floor plan because it looks nicer and convenient. Next, I also do parts that needs explanation such as at task 6 and task 8. I also give ideas and opinion during this project discussion that is used in the final result.

These are the floor plans that we came out with:



Next, all my group members have done their best in completing the task given in this project, first is Mohamad Fakhurulradhi bin Mohd Sha'ri, as a group leader, he is very determined in doing every task, he explains all the question asked and divide the work that need to be done. He is also the one that compile all of our work into one neat report using the appropriate format. We are very grateful to have a leader like him.

After that, Tiffanny Anak Garan, she also done a very good job during this project. When it comes to calculating the cost and budget for the project, she is the one that does the calculation. She also contribute by giving a lot of ideas and she also helps me in doing explanations for some question.

Then, Natalie Nuai Anak Gerald, she also contributes to the group very well. She has do the drawing of the topology for this project and also did the drawing for task 7 along with me. She also gives a lot of useful ideas and opinion trough out our group meeting.

We work as a group by doing a meeting whenever there are new task given. For example, if the task requires a research on the internet, we will go to our faculty in order to get access to a faster internet because with faster internet, our job will be easier and the task given will be done in time.

We also work as a group by dividing the work in every task because we want to make sure that all group members contribute and give their ideas in making this project a success.

Finally, my overall comments on this project is that it will very useful and logical for future use, the internet connection will be fast by looking at the devices that we decided to use especially the fiber optic part. Even though it is quite exhausting, we all work as a team in completing this project and the time spent during our meeting is very fun and I hope that other team members feel the same way.



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SEMESTER II – 2014/2015

SCSR1213-08

Network Communications

Personal Report

Name : Natalie Nuai Anak Gerald
Matrix Number : A14CS0085
IC Number : 950514-13-6116
Year programme : 1 SCSV
Section : 08
Lecturer Name : En. Muhalim Bin Mohamed Amin

Personal Report

My group name is “fantastic_4”. Our group members is me, Natalie Nuai Anak Gerald, Tiffany Anak Garan, Syakir Zufayri Bin Roslan and Mohamad Fakhrulradhi Bin Mohd Sha’ri. I’m glad to have this people in my group because all of us were really doing our works. We know when is the time to play and when the time to be serious. When it’s come to works, we are very serious to discuss about anything in this project as we were given five hundred thousand for this project. From the starting point, we have plan very well before continue to any tasks.

What I contribute in this group is I’m more doing in sketching and measuring. I’m using “Cisco Packet Tracer Student” software for some of the part of the sketching. The software was easy to use which make me faster doing my works. I became familiar to use this software because we had been using this software when doing lab in the class. Besides that, I’m also using “Microsoft Word” to do the sketching part. It’s easier to make the shape, line, and to add some words. But still all of us are not specific any works, I also did some of the essays. In addition, Syakir also do some of the part sketching. Both of us had divided the sketch part.

The other members have done very perfect job. As I mention above, Syakir also do some of the part sketching. On contrary, Syakir also do the essays part. While other group members, Tiffany, do calculation for the price. The four of us had been searching the devices and choose the cheaper price to use for our projects. We choose cheaper price, but we still do consider the condition. If the device was not so good even though it was cheap, still we choose the most advantage device even it’s expensive. This is because our objective are consider about the future. We do not want to waste our money to buy the same devices every year. Lastly, Tiffany compile all the devices and the best price we want to use. Other than that, Tiffany also do the essay part. Whereas Radhi, dividing our tasks and explain to three of us what to do, explain about anything in the project. For example, some part of the network we do not understand, so Radhi explain it. If four of us not understand, so we use internet to search about it. Then, we will discuss about it to make it clearer. Radhi mostly do the compilation all of our works. Nevertheless, he was doing a correction for anything we had done wrongly. Besides that, Radhi also do the essay part. We help each other to make it faster to complete. What to say, our management was very perfect.

For every task we were doing it together either at “KTDI meeting room” or at faculty. Our management throughout this project is very efficient which is we split our tasks so it will save time. Everyone have their own part to do. When we were doing it together, we will finish it at that time no matter what. We do not want to waste our time. So we were being very serious while doing the tasks. But, some maybe could not complete on that time, so we continue do at our room. What I remember is when we were doing task 5 and 6, we doing it at faculty because easier for us to connect with internet. We started our discussion at 9.30pm until 5.00am. That was really my first time sleep that late. All of our members was so tired, but glad to say that we could finish it before 5.00am because we thought that submission must be done before 5.00am. Although all of us tired, still we could see the smile of each other as we done our tasks.

This project is not easy as we thought. Although some of the tasks were very difficult, thus we manage to do all the 10 tasks very well. Of course I’m proud of my teammate because all of us had done the very best we could. All the hardship we have been through while doing this tasks has made us more matured. Even we stayed up all night just to do the tasks still we satisfied with what we doing. Well, it’s part of student life. On the other hand, we become more responsible and can get more experience of course. This project also taught us how to manage time very well. We could submit all the tasks on time which is we are very punctual. This project was really a high level which was a very good project for student. This is because we could apply what we study in the lectures. This project taught us how to handle a project with a lot amount of money. Of course, it will be very useful when we working someday.

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